

Teacher Guide — middle / module-06- moon-phases-tithi

IKS Curriculum

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Overview

Module 6 — Moon Phases & Tithi

Age band: Middle · **Grades:** 6–8 · **Duration:** 10 sessions × 45 min (≈ 2 weeks)

Learning objectives

By the end of this module, students will be able to:

1. **Describe** the synodic month (~29.5 days) and explain why the Moon appears to change shape each night using the Sun–Moon–Earth geometry.
2. **Define** *tithi* as one-thirtieth of a synodic month and explain why a tithi need not equal a calendar day.
3. **Name** the two *pakṣas* (*śukla* and *kṛṣṇa*) and the 15 tithis of each, recognising the Sanskrit ordinal pattern from *pratipadā* to *caturdaśī* plus *pūrṇimā* and *amāvasyā*.
4. **Build** a tithi calendar for the current calendar month using a published *pañcāṅga* or astronomy app.
5. **Explain** why solar eclipses occur on *amāvasyā* and lunar eclipses on *pūrṇimā*, and why they do not happen every month.
6. **Compare** the lunisolar Indian calendar with three other lunar/lunisolar calendars (Islamic Hijri, Chinese, Hebrew) and identify what is shared and what differs.

NEP 2020 alignment

- **§4.6** — Integration of Indian Knowledge Systems with mainstream astronomy.
- **§4.23** — Experiential learning through nightly Moon observation and torch-and-ball modelling.
- **§11.8** — Multidisciplinary linkage (astronomy + Sanskrit + calendar systems + history).
- **§4.27** — Local context: students build a calendar for their own month and link it to local festivals.

Prerequisites

- Familiarity with the day-night cycle and the Earth’s rotation (NCERT Class 6 Science).

- Comfort with degrees and basic geometry (a full circle is 360°).
- No Sanskrit knowledge assumed; all terms introduced with IAST + meaning.

Key vocabulary

See `_shared/glossary.md` for full entries.

- *tithi* — lunar day, 1/30 of a synodic month
- *pakṣa* — fortnight; *śukla* (waxing) and *kṛṣṇa* (waning)
- *amāvasyā* — new moon; *pūrṇimā* — full moon
- *pañcāṅga* — five-limbed almanac: tithi, vāra, nakṣatra, yoga, karaṇa
- *nakṣatra* — one of 27 lunar mansions
- *adhika māsa* — intercalary “extra” month

Materials checklist (whole module)

- A torch (flashlight) and a small white ball (table-tennis ball works) for the lighting demo
- A printed lunar phase calendar for the current month (NASA publishes monthly phase charts; any Indian *pañcāṅga* app also works)
- Chart paper for the class tithi calendar
- Index cards (~30) for the tithi sorting activity
- A globe and a smaller spherical object (for eclipse geometry)
- Project supplies (poster paper, optional camera or smartphone)

Day-by-day overview

Day	Title	Focus	Assessment
1	The Moon and Why It Changes	Sun–Moon–Earth geometry; torch-and-ball demo	—
2	The Two Pakṣas	Waxing and waning fortnights	Exit ticket
3	Naming the 15 Tithis	Sanskrit ordinals; tithi vs. day	Formative quiz
4	Build a Tithi Calendar	Apply to current calendar month	Activity rubric
5	Lunar Months and Festivals	Mid-module check; festival mapping	Formative quiz
6	The Pañcāṅga — Five Limbs	Tithi, vāra, nakṣatra, yoga, karaṇa overview	Reflection
7	Eclipses — Why and When	Solar and lunar geometry; safety	Discussion
8	Project Session 1	Plan + first artefact	Project draft

Day	Title	Focus	Assessment
9	Cross-cultural Lunar Calendars	Hijri, Chinese, Hebrew + project build	Project draft
10	Final Assessment + Showcase	Summative quiz + 3-min presentations	Summative + project

Citation sources used

- **Modern astronomy:** NCERT Physics Class XI, Chapter 8 (Gravitation) for orbital motion; NASA’s published lunar phase data for current-month verification.
- **Indian sources:** *Sūrya-siddhānta* (compiled c. 5th–10th c. CE) — chapter 2 (lunar motion) referenced for the tithi computation rule; paraphrased only at Middle band.
- See `sources.md` for the full list.

Project brief (one-line teaser)

Each student (or pair) keeps a 10-night Moon observation log, builds a tithi calendar for one calendar month, and produces a poster, slideshow, or short video explaining one lunar phenomenon — phases, eclipses, or the intercalary month — using both the *pañcāṅga* and modern astronomy. Full brief in `assessments/project-brief.md`.

10-Day Lesson Plan

Module 6 (Middle) — 10-Day Lesson Plan

At a glance

Day	Session title	Lesson file	Activity / assessment
1	The Moon and Why It Changes	<code>days/day-01-moon-and-why.md</code>	Torch-and-ball demo
2	The Two Pakṣas	<code>days/day-02-paksha.md</code>	Diagram-build, exit ticket
3	Naming the 15 Tithis	<code>days/day-03-fifteen-tithis.md</code>	Tithi-ordinal drill; formative quiz
4	Build a Tithi Calendar	<code>days/day-04-tithi-calendar.md</code>	<code>activities/activity-01-build-tithi-calendar.md</code>
5	Lunar Months and Festivals	<code>days/day-05-lunar-months.md</code>	<code>quizzes/formative.md</code> (Part B)
6	The Pañcāṅga — Five	<code>days/day-06-</code>	Compare-contrast worksheet

Day	Session title	Lesson file	Activity / assessment
	Limbs	panchanga.md	
7	Eclipses — Why and When	days/day-07-eclipses.md	activities/activity-02-eclipse-geometry.md
8	Project Session 1: Plan	days/day-08-project-session-1.md	Project draft submitted
9	Cross-cultural Lunar Calendars + Project Build	days/day-09-cross-cultural.md	Project draft submitted
10	Final Assessment + Showcase	days/day-10-final-assessment.md	quizzes/summative.md + project

Pacing notes

- Days 1–3 are content-heavy. If your class needs a half-day buffer, split Day 3 across two sessions and roll Days 5–10 forward by one slot. Drop Day 9’s cross-cultural segment if you must compress; assign it as reading.
- Day 4’s tithi calendar benefits from advance setup — print one *pañcāṅga* page (or open an astronomy app) for the current month before class begins.
- The 10-night Moon observation diary (homework) starts on Day 1 and runs across the module. Remind students every day for the first three days.
- Project work (Days 8–9) assumes you’ve decided pair vs. individual by Day 5.

Suggested running theme

Open each class with the same question on the board: **“Did you see the Moon last night? What shape was it?”** Two students share for 30 seconds each. By Day 10 every student will have spoken at least twice. This builds the observation habit the module is designed to grow.

Formative vs. summative weight

- Formative quizzes (Days 3 and 5) are **not graded** — they are diagnostic. Use them to decide who needs scaffolding on Day 6.
- Summative quiz (Day 10) — see rubric in `assessments/rubric.md`. Combined with project, total 40 marks.

What students leave the module with

- A 10-night Moon-shape diary in their notebook with tithi names.
- A completed tithi calendar for one calendar month.
- The 15 tithi names with English translation, in their own handwriting.
- One completed project artefact.
- The habit of looking up — and noticing the Moon — most nights.

Teacher Notes

Teacher Notes — Module 6 Moon Phases & Tithi (Middle)

Posture for this module

Lead with what students can see — the Moon, tonight, in the sky. The *tithi* system is a sophisticated organisation of those observations, refined by Indian astronomers over many centuries. It is not folklore. It is also not modern astronomy. It is a calendrical scheme that does some things modern astronomy does not bother with (a unit smaller than a day, tied to the actual position of the Moon) and skips some things modern astronomy demands (a fixed integer number of days per month).

Treat the *Sūrya-siddhānta* as a serious astronomical text — one of several in the world’s history of astronomy — not as a mythological document. If a student asks “is this religious?” — the honest answer is that the *pañcāṅga* is used both for religious timing (festival dates) and for civil/agricultural timing. We use it here as a calendrical and astronomical instrument; the religious uses are real and respected but not our focus.

Common misconceptions to watch for

1. **“A tithi is one day.”** A tithi is 1/30 of a synodic month, not 1/30 of a calendar day. Average length ~23.6 hours, but can range from ~21.5 to ~26 hours because the Moon’s motion is non-uniform. A given tithi may “skip” a sunrise (*kṣaya*) or span two sunrises (*vrddhi*).
2. **“The new moon means no Moon.”** The Moon is still there during *amāvasyā*. It’s just between the Sun and Earth from our perspective, so the illuminated half faces away from us.
3. **“Phases happen because of Earth’s shadow.”** A very common confusion. Phases happen because the Sun lights up only half of the Moon at any time, and we see different fractions of that lit half as the Moon orbits Earth. Earth’s shadow causes only lunar *eclipses*, not phases.
4. **“Eclipses happen every new moon and full moon.”** They would, if the Moon’s orbit were in the same plane as Earth’s orbit around the Sun. The Moon’s orbit is tilted ~5°, so eclipses happen only when the Moon crosses through this plane near new or full moon. About 2–5 solar eclipses and 0–3 lunar eclipses per year.
5. **“All Indian calendars start on the same date.”** No — *amānta* (month ends at new moon, used in much of South and Western India) and *pūrṇimānta* (month ends at full moon, used in much of North India) traditions number the months differently. The same tithi can belong to different months under different conventions.
6. **“The śukla pakṣa is the ‘good’ fortnight.”** Both pakṣas are simply astronomical halves. Festivals and rituals are tied to both — *amāvasyā* (kṛṣṇa pakṣa) is auspicious for ancestor rites; *pūrṇimā* (end of śukla pakṣa) is auspicious for other observances.

Safety — eclipse viewing

This is the most important safety note in the module:

- **NEVER look directly at a solar eclipse**, even partial. Looking even briefly without proper protection can cause permanent retinal damage. Ordinary sunglasses do not work. Smoked glass does not work. Welder's glass below shade #14 does not work.
- **Safe methods:** ISO 12312-2 certified eclipse glasses, pinhole projection (project the Sun's image onto a sheet of paper through a small hole), or telescope projection (never look through the telescope — project the image).
- **Lunar eclipses are safe to watch directly.** They are dim — sometimes the Moon turns coppery-red ("blood moon") because the only light reaching it is sunlight bent through Earth's atmosphere.

State this safety rule at least twice in Day 7. Make every student write it in their workbook.

On Sanskrit pronunciation

Drill these terms aloud in week 1; students will refer to them throughout the module:

- *tithi* — "TI-thi" (short *i* both times)
- *pakṣa* — "PUK-sha" (the ṣ is a retroflex sh)
- *śukla* — "SHU-kla"
- *kṛṣṇa* — "KRISH-na" (the ṛ is a vowel)
- *amāvasyā* — "ah-MAH-vahs-yah"
- *pūrṇimā* — "POOR-ni-mah"
- *pañcāṅga* — "PUN-chahn-ga"
- *nakṣatra* — "NUCK-shu-tra"

Religious / secular framing

Some families observe fasts (*vrata*) on specific tithis — *ekādaśī* (11th), *caturdaśī* (14th), *pradoṣa* (13th evening), etc. A student may volunteer this. Acknowledge it briefly — "yes, *that's a traditional practice tied to this tithi*" — and return to the astronomical content. Do not lead with the religious dimension.

If a student asks why eclipses are "considered inauspicious" — answer historically: most cultures across the world treated eclipses as significant events, often inauspicious, because they were rare and unpredicted in pre-scientific contexts. Today we understand the geometry and can predict eclipses years in advance; the astronomical fact is decoupled from the cultural interpretation.

Differentiation hints

- **Tier up (fast finishers):** Ask them to compute the tithi at a specific time on a specific date using the elongation formula ($\text{tithi} = \text{floor}((\text{Moon_longitude} - \text{Sun_longitude}) \bmod 360 / 12) + 1$). They will need an astronomy app for the longitudes; this is genuinely how the modern *pañcāṅga* is computed.

- **Tier down (scaffolding):** Reduce the first lesson to “the Moon changes shape because the Sun lights it up from different angles” — defer the precise geometry to Day 7’s diagram. Memorisation of all 15 tithi names can wait until Day 5.

Sensitivity notes

- Some students may come from families who consult a *pañcāṅga* daily for muhūrta (auspicious timing). Honour that without privileging it. The framing is “this is an astronomical instrument that has these other uses in the tradition.”
- Avoid implying that Indian astronomy “anticipated” or “was ahead of” Western astronomy. The histories are parallel; both made remarkable observations; both made errors. Treat the *Sūrya-siddhānta* as an astronomical text on its merits.

Project oversight (Days 8–10)

- Confirm groups/pairs by end of Day 5.
- On Day 8, every group should leave class with a one-paragraph plan and a chosen lunar phenomenon (phases / eclipses / intercalary month).
- On Day 9, intervene if a group is still planning — they need a draft artefact by end of class.
- For Day 10 presentations, use the 3-minute timer strictly. The discipline of compression is part of the learning.

If a clear sky is unavailable for the observation diary

This is a real risk during monsoon or in light-polluted cities. Allow students to use: - A NASA-published lunar phase image for the date. - An astronomy app (Stellarium, SkyView, etc.) showing the Moon’s phase for the date. - A reliable online phase calendar.

State up front that the diary’s purpose is the *habit of looking up*. If they can’t see the Moon, they should still attempt — note “cloudy, could not see” and the expected phase from a published source.

Assessment grading sequence

1. Mark Day 5 formative quiz immediately so feedback lands within 24 hours.
2. Day 10 summative quiz can be marked over the weekend.
3. Project rubric should be discussed *with* students at the start of Day 8.

If you only have 6 days, not 10

Compress as follows: - Day 1 (was Day 1): Moon + torch-and-ball - Day 2 (was Day 2 + 3 condensed): Pakṣa + tithi names - Day 3 (was Day 4): Build the tithi calendar - Day 4 (was Day 7): Eclipses + safety - Day 5 (was Day 8 + 9): Project work - Day 6 (was Day 10): Summative + showcase

Skip Day 5 mid-module check and Day 9 cross-cultural in this compressed version.

Sources

Sources — Module 6 Moon Phases & Tithi (Middle)

Indian astronomical sources (paraphrased; no direct quotation at Middle band)

Source	Used in	Content
Sūrya-siddhānta , chapter 2 (lunar motion)	Day 3, Day 6, Day 7	The text defines the tithi as the interval during which the Moon gains 12° of elongation on the Sun. It gives algorithms for tithi length, eclipse prediction, and <i>adhika māsa</i> insertion. We paraphrase the rule; we do not quote any single verse.
Pañcasiddhāntikā of Varāhamihira (6th c. CE)	Day 6 (teacher background)	Summarises five earlier siddhāntas; useful context that tithi computation predates the Common Era. Not student-facing.
Traditional pañcāṅga publications	Day 4, Day 5, Day 9	Any current-year pañcāṅga (printed or app — Drik Panchang, Mypanchang.com, etc.) provides tithi values for verification of the class calendar. Choose one and stick to it across the module.

No invented citations. This module does not quote any specific verse number from the *Sūrya-siddhānta* or *Pañcasiddhāntikā* because we cannot verify a single chapter-and-verse reference cleanly without secondary scholarship. Paraphrase is the safe move at Middle band; the Senior band engages with the texts in their own way.

Modern astronomy references

These are standard middle-school astronomy and are recommended for cross-checking:

- **NCERT Physics Class XI**, Chapter 8 (“Gravitation”) — for the orbital motion of the Moon around Earth.
- **NCERT Science Class 6**, Chapter 17 (“Stars and the Solar System”) — for the basic Sun–Moon–Earth geometry.

- **NASA’s published lunar phase data** — visit the NASA Solar System Exploration site and search for “Moon phases [current year]” for an official phase calendar to compare against. (Do not invent specific URLs; tell students to search the official NASA site.)
- **In The Sky** astronomical reference (in-the-sky.org) — a verified secondary source for upcoming eclipses, used for Day 7.

Cross-cultural calendar references (Day 9)

- **Islamic (Hijri) calendar** — strictly lunar; 12 months $\times$ ~ 29.5 days = ~ 354 days. No intercalation. Months drift through the seasons over ~ 33 years. See any standard reference; the *Encyclopaedia of Islam* entry on *taqwīm* is comprehensive.
- **Chinese calendar** — lunisolar with intercalary months, similar in principle to the Indian system. Lunar New Year falls between late January and mid-February.
- **Hebrew calendar** — lunisolar with the Metonic cycle (7 intercalary months in 19 years).

Historical context references (teacher background)

- Plofker, K. (2009). *Mathematics in India*. Princeton University Press. — Sets context for the *Sūrya-siddhānta* tradition and Indian planetary astronomy in chapter 4.
- Pingree, D. (1981). *Jyotiḥśāstra: Astral and Mathematical Literature*. Otto Harrassowitz. — Standard reference on the astronomical-astrological literature; teacher reference only.

(These are teacher references; not assigned to students at Middle band.)

RAG note

No RAG database was used for this module. The *vidya-karana-kg* RAG does not cover astronomical literature in the current build, so we rely on general published references and paraphrase the *Sūrya-siddhānta* rules.

Day 1 — The Moon and Why It Changes

Module: Moon Phases & Tithi · **Band:** Middle · **Time:** 45 min

Learning objective

Students will explain, with a physical model, why the Moon appears to change shape over a ~ 29.5 -day cycle, and will name the synodic month as the underlying period.

Materials

- A torch / flashlight (the brighter the better — bring spare batteries)
- A small white ball (a table-tennis ball or a styrofoam ball about 4 cm across)
- A darkened classroom (curtains drawn or lights off for ~ 10 min)
- Whiteboard
- Notebooks

Vocabulary introduced today

- *synodic month* — the ~29.53-day period from one new moon to the next; the basis of the lunar calendar.

(Sanskrit terms start tomorrow. Day 1 is observational and physical.)

Lesson flow

Warm-up (5 min)

- Opening question on board: “*Did you see the Moon last night? What shape was it?*”
- Take 4–5 students’ answers. Sketch each shape they describe on the board (don’t correct yet — just record).
- “*Why is it sometimes a full circle, sometimes a thin sliver, sometimes nothing at all? That’s what this module answers.*”

Core (20 min)

1. **The Moon is a ball lit by the Sun.** Hold up the white ball. “*This is the Moon — a rocky sphere, about a quarter the size of the Earth.*” Shine the torch on it from one side. “*The Sun lights up only one half of the Moon. Always. Half is day, half is night.*”
2. **Why the shape changes — the demo.** Have one student stand still in the centre of the room (= Earth). One student stands ~3 m away holding the torch on (= Sun). You walk slowly around the Earth-student in a wide circle, holding the ball at arm’s length so the Earth-student can see it.

At each of 8 stopping points, ask: “*What fraction of the lit half of the ball can Earth see?*”

- **Position 1 (Moon between Earth and Sun):** Earth sees the dark side. → *new moon*
 - **Position 2 (45° on):** A thin sliver visible on one side. → *waxing crescent*
 - **Position 3 (90°):** Half of the disc visible. → *first quarter*
 - **Position 4 (135°):** Most of the disc visible. → *waxing gibbous*
 - **Position 5 (Earth between Sun and Moon):** Earth sees the fully lit side. → *full moon*
 - **Positions 6–8:** Reverse — gibbous, half, crescent → *waning*
3. **Key insight on the board:**
The Moon doesn’t change. The angle from which we see it changes.
 4. **The cycle takes ~29.5 days.** This is the **synodic month** — the time from one new moon to the next. Write the number 29.53 on the board.
 5. **One question to plant:** “*Earth orbits the Sun in 365 days. The Moon orbits Earth in 29.5 days. Is the lunar ‘month’ the same as a calendar month?*” (No — and that’s why we have leap years for the solar calendar, and a fancier system for the lunar one. We’ll get to that on Day 5.)

Activity (15 min) — Pair sketching

In pairs:

1. One partner is “Sun” (holds an imaginary torch in one direction).
2. The other is “Earth” (stays still).
3. Take a small ball or a balled-up paper and walk it around Earth in 8 stopping positions.
4. At each position, the Earth-partner sketches what shape they see in their notebook.

After 7 minutes, swap roles.

By end of activity, each student has 8 small Moon-shape sketches in their notebook labelled new, waxing crescent, first quarter, waxing gibbous, full, waning gibbous, last quarter, waning crescent.

Wrap-up (5 min)

- **Exit ticket on a slip of paper, 2 min:** “*In one sentence: why does the Moon look different on different nights?*”
- Collect. Read 3 aloud quickly. Correct any “Earth’s shadow makes the phases” answers gently — Earth’s shadow only matters for *eclipses* (Day 7).
- **Tonight’s homework:** Look at the Moon. Note the date, the time, the shape (sketch). Write down the cloud condition. This starts your **10-night Moon diary**.

Student-facing content

The Moon does not change. Our view of it changes.

The Sun lights up half of the Moon. As the Moon orbits Earth, we see different fractions of that lit half. Over ~29.5 days, the visible fraction grows from nothing (new moon), to a half-disc (quarter), to a full disc (full moon), back to a half, and back to nothing.

This 29.5-day cycle is called the **synodic month** — the basis of the lunar calendar that we’ll study in this module.

Homework

- **Tonight (and every night for 10 nights):** Look at the Moon. Sketch its shape in your workbook. Note the date, the time, and “clear / partly cloudy / overcast.” If overcast, look it up on a Moon-phase app or NASA’s published phase calendar.

Differentiation

- **One tier up:** Why is the synodic month (29.53 days) different from the *sidereal* month (27.32 days, the time for the Moon to return to the same star background)? Hint: Earth has moved around the Sun during the lunar month.
- **One tier down:** Skip the 8-position drill. Focus on 4: new, half (waxing), full, half (waning). Build the rest later in the week.

Teacher notes

- Some students will say “the Moon makes its own light.” It does not — it reflects sunlight. Show them this physically by switching the torch off in the demo. The ball goes dark.
- Some will say “the Moon’s far side is permanently dark.” It is not. The Moon’s far side gets just as much sunlight as the near side, on average — we just never see the far side from Earth because the Moon is tidally locked (the same face always faces us).
- For the diary, if students live in a heavily light-polluted city, the thin crescents may be hard to spot in twilight. They can use an app for verification — that’s acceptable.

Citation(s) used in this lesson

- General reference: NCERT Science Class 6, Chapter 17 (“Stars and the Solar System”). No verse citation today.

Day 2 — The Two Pakṣas

Module: Moon Phases & Tithi · **Band:** Middle · **Time:** 45 min

Learning objective

Students will define *śukla pakṣa* and *kṛṣṇa pakṣa*, identify the two anchor days *amāvasyā* and *pūrṇimā*, and place yesterday’s observed Moon shape into one of the two pakṣas.

Materials

- Yesterday’s exit-ticket slips (skim before class)
- Board with two columns ready: *śukla pakṣa* and *kṛṣṇa pakṣa*
- The white ball and torch (for a 2-min recall demo)
- Notebooks

Vocabulary introduced today

- *pakṣa* (IAST: pakṣa; lit. “wing, side, fortnight”) — half of a lunar month. There are exactly two pakṣas in every lunar month.
- *śukla pakṣa* (IAST: śukla-pakṣa; lit. “bright fortnight”) — the waxing half: the Moon grows from invisible to full.
- *kṛṣṇa pakṣa* (IAST: kṛṣṇa-pakṣa; lit. “dark fortnight”) — the waning half: the Moon shrinks from full back to invisible.
- *amāvasyā* (IAST: amāvasyā; lit. “dwelling-together”) — new moon; the Moon and Sun are in the same direction in the sky.
- *pūrṇimā* (IAST: pūrṇimā; lit. “fullness”) — full moon; Moon directly opposite the Sun.

Lesson flow

Warm-up (5 min)

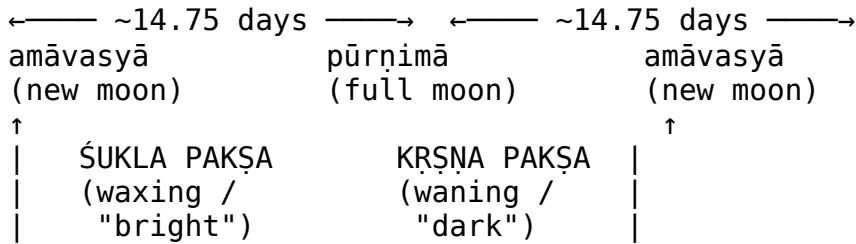
- Daily prompt: “Did you see the Moon last night? What shape?” — 2 students share.

- Quick recall: ring the question from yesterday’s demo. “*Why does the Moon change shape?*” (Sun lights half; we see different fractions as it orbits.)
- One-sentence reminder: “*The full cycle takes about 29.5 days — the synodic month.*”

Core (20 min)

1. Split the cycle in half.

Draw on the board:



- The fortnight from *amāvasyā* to *pūrṇimā* — when the Moon is **growing** — is the *śukla pakṣa* (“bright fortnight”). The visible lit fraction is increasing each night.
 - The fortnight from *pūrṇimā* to *amāvasyā* — when the Moon is **shrinking** — is the *kṛṣṇa pakṣa* (“dark fortnight”). The visible lit fraction is decreasing each night.
2. **Why these names?** *Śukla* (शुक्ल) means “bright, white”; *kṛṣṇa* (कृष्ण) means “dark, black.” Naming the fortnights by the direction the brightness is heading — toward full or toward new — is precise and useful.
 3. **A common confusion to head off:** Neither pakṣa is “good” or “bad.” Both are simply astronomical halves of the lunar cycle. Some festivals fall in *śukla* (e.g. Rāma Navamī, Vaikuṇṭha Ekādaśī), some in *kṛṣṇa* (e.g. Janmāṣṭamī, Mahā-śivarātri).
 4. **The two anchor days deserve their names:**
 - *amāvasyā* — literally “dwelling-together.” The Sun and Moon are in the same direction in the sky (Moon is between Earth and Sun-ish, near the same celestial longitude). The Moon is up during the day, invisible against the sky’s brightness.
 - *pūrṇimā* — “fullness.” Moon is opposite the Sun. It rises as the Sun sets, stays up all night, sets as the Sun rises. It is fully illuminated.
 5. **Connecting to yesterday’s diary.** Hold up your own diary — “*I observed last night, it was a thin crescent right after sunset on the western horizon. That’s a waxing crescent — I’m in śukla pakṣa, just a few days after amāvasyā.*”

Have 3 students share theirs: which pakṣa, how do they know?

Activity (15 min) — Sort the shapes

On the board, draw 8 Moon shapes (you sketched these yesterday in the demo). Number them 1–8 in random order.

In pairs, students:

1. Copy the 8 shapes (small, in a row) in their notebook.
2. Label each as *śukla* or *kṛṣṇa* — and give a reason.
3. Number them in the order they would appear across a lunar month, starting from *amāvasyā*.

Discuss with a different pair (3 min): did you disagree on any? Which?

Wrap-up (5 min)

- **Exit ticket (1 min):** “What is the difference between *śukla pakṣa* and *kṛṣṇa pakṣa*? Use one sentence.”
- Preview Day 3: “Tomorrow we name the 15 days within each *pakṣa* — the *tithis*.”

Student-facing content

A lunar month has two halves — two *pakṣas*.

Sanskrit	English	Direction the Moon is heading
<i>śukla pakṣa</i>	bright fortnight	growing — new moon → full moon
<i>kṛṣṇa pakṣa</i>	dark fortnight	shrinking — full moon → new moon

Two anchor days bookend the *pakṣas*: - *amāvasyā* (new moon) — Moon invisible, “dwelling together with” the Sun - *pūrṇimā* (full moon) — Moon fully lit, opposite the Sun

Neither *pakṣa* is “good” or “bad.” They are astronomical halves of the cycle.

Homework

- Continue the Moon diary. Tonight, add **which *pakṣa* you are in** to your sketch.
- For Day 3, **memorise the names *amāvasyā* and *pūrṇimā* and the two *pakṣa* names.**

Differentiation

- **One tier up:** Look up the meaning of the words *śukla* and *kṛṣṇa* in any Sanskrit dictionary. Are they used only for the Moon, or in other contexts? (They are general-purpose words for “bright/white” and “dark/black.”)
- **One tier down:** Memorise only *amāvasyā* and *pūrṇimā* for now. The *pakṣa* terms can wait one more day.

Teacher notes

- Watch for the misconception that *kṛṣṇa pakṣa* is “the dark side of the Moon” — these are completely different concepts. *Kṛṣṇa pakṣa* is a fortnight of the calendar; the “far side” of the Moon is a separate thing about which face we always see.

- Pronunciation hint: the ṣ in *pakṣa* is retroflex — closer to English “puk-sha” than “puk-sa.” Don’t worry about perfection; insist on “pakṣa,” not “paksha-aaa.”

Citation(s) used in this lesson

- Paraphrase from *Sūrya-siddhānta*, chapter 2 (lunar motion): the lunar month is divided into 30 tithis, distributed 15 each across the two pakṣas. Cited by text name only.

Day 3 — Naming the 15 Tithis

Module: Moon Phases & Tithi · **Band:** Middle · **Time:** 45 min

Learning objective

Students will define *tithi* in terms of the Moon’s elongation from the Sun (12°), recite the 15 tithi names of a pakṣa in order, and explain why a tithi need not equal a 24-hour calendar day.

Materials

- Board with three columns ready: Number | IAST name | English (ordinal)
- A printed handout of the 15 tithi names (one per student)
- The white ball and torch from Day 1 (one quick re-use)
- Notebooks

Vocabulary introduced today

- *tithi* (IAST: tithi; lit. “lunar day”) — $1/30$ of a synodic month; precisely, the time the Moon takes to gain 12° of angular distance (“elongation”) from the Sun.
- *elongation* — the angle between the Sun and Moon as seen from Earth (0° at *amāvasyā*; 180° at *pūrṇimā*).

Lesson flow

Warm-up (5 min)

- Daily prompt: “*Did you see the Moon last night? What pakṣa is it?*” — 2 students share.
- Recall: write on the board — “29.5 days, 2 pakṣas, 2 anchor days.” Ask the class to fill in: how many days in a pakṣa? (~ 14.75 .) How many parts will we divide that into? (15.)

Core (20 min)

1. The tithi as a unit.

“A tithi is one-thirtieth of a full lunar cycle. So a synodic month has 30 tithis. Each pakṣa has 15.”

On the board:

Synodic month (29.53 days)

└ Śukla pakṣa = 15 tithis (1 → 15)
└ Kṛṣṇa pakṣa = 15 tithis (1 → 15)

2. **The precise definition (this is important).**

“A tithi is NOT one-thirtieth of a day. It is one-thirtieth of the cycle.”

The full cycle covers 360° of elongation (the Moon goes all the way around). Divide 360° by 30 — each tithi is 12° of elongation.

One tithi = the time during which the Moon gains 12° of elongation on the Sun.

On average, this takes about 23 hours and 37 minutes. But the Moon doesn’t move uniformly — sometimes faster (perigee, closer to Earth) and sometimes slower (apogee, farther). So a single tithi can last anywhere from ~21.5 hours to ~26 hours.

3. **Why a tithi is not a day — three consequences.**

- Some tithis “skip” a sunrise. If a tithi starts at 7 a.m. and ends at 5 a.m. the next morning, it never overlaps with a sunrise. This is called *kṣaya* tithi (“decayed” — it never had a sunrise of its own).
- Some tithis “span” two sunrises. If a tithi starts at 5 a.m. on one day and ends at 7 a.m. two days later, it covers two sunrises. This is *vṛddhi* (“growth”).
- Most tithis are tagged to one calendar date — the date during which the tithi is current at sunrise.

4. **The 15 names — they are Sanskrit ordinals.**

On the board, build the list. Students copy. (Show *śukla* column; same names apply to *kṛṣṇa* with the prefix.)

#	IAST	English (ordinal)
1	<i>pratipadā</i>	“first”
2	<i>dvitīyā</i>	“second”
3	<i>tṛtīyā</i>	“third”
4	<i>caturthī</i>	“fourth”
5	<i>pañcamī</i>	“fifth”
6	<i>ṣaṣṭhī</i>	“sixth”
7	<i>saptamī</i>	“seventh”
8	<i>aṣṭamī</i>	“eighth”
9	<i>navamī</i>	“ninth”
10	<i>daśamī</i>	“tenth”
11	<i>ekādaśī</i>	“eleventh”
12	<i>dvādaśī</i>	“twelfth”
13	<i>trayodaśī</i>	“thirteenth”

#	IAST	English (ordinal)
14	<i>caturdaśī</i>	“fourteenth”
15	<i>pūrṇimā</i> (or <i>amāvasyā</i>)	full / new moon

Note the pattern: 1–10 are simple ordinals; 11–14 are compound (*eka+daśī*, *dvā+daśī*, *trayaḥ+daśī*, *catur+daśī*); the 15th is always the anchor day — *pūrṇimā* at the end of *śukla*, *amāvasyā* at the end of *kṛṣṇa*.

5. **Sanskrit number recall.** *Eka*, *dvi*, *tri*, *catur*, *pañca*, *ṣaṣ*, *sapta*, *aṣṭa*, *nava*, *daśa* — count it together once. The tithi names will stick faster if students see them as ordinals first, Sanskrit terms second.

Activity (15 min) — Tithi drill

Two parts:

Part A (8 min) — paired recall. In pairs, partners alternate: one says a tithi number (1–15), the other names the tithi in IAST. Switch roles every minute. Goal: name all 15 in under 60 seconds by end of activity.

Part B (7 min) — date-to-tithi. Distribute a printed list of 5 dates and the tithi at sunrise on each (lookup done by the teacher from any standard *pañcāṅga* — e.g. for a sample week). Students answer: - *Is this date in śukla or kṛṣṇa pakṣa?* - *What’s the Sanskrit name of the tithi?* - *What shape will the Moon be that night?*

Walk around. Check that students are reading the table, not guessing.

Wrap-up (5 min)

- **Formative quiz on a slip of paper (5 questions, 3 min)** — see quizzes/formative.md Part A.
- Collect for grading. Next day’s class includes the result.
- **Preview Day 4:** “Tomorrow we build a tithi calendar for THIS calendar month.”

Student-facing content

A tithi is the time the Moon takes to gain 12° on the Sun.

- $360^\circ / 30 = 12^\circ$ per tithi.
- Average tithi length: ~23 h 37 min — close to a day but not exactly a day.
- Some tithis skip a sunrise (*kṣaya*); some span two (*vṛddhi*).

Each pakṣa has 15 tithis named as Sanskrit ordinals:

pratipadā, *dvitīyā*, *ṛtīyā*, *caturthī*, *pañcamī*, *ṣaṣṭhī*, *saptamī*, *aṣṭamī*, *navamī*, *daśamī*, *ekādaśī*, *dvādaśī*, *trayodaśī*, *caturdaśī*, *pūrṇimā* (or *amāvasyā*).

Homework

- Memorise the 15 tithi names. Quiz yourself or be quizzed at home.
- Continue the Moon diary — now naming the tithi as well as the pakṣa.

Differentiation

- **One tier up:** Look up the formula for tithi computation: $\text{tithi number} = \text{floor}((\text{Moon_ecliptic_longitude} - \text{Sun_ecliptic_longitude}) \bmod 360) / 12) + 1$. Use an astronomy app (Stellarium is free) to get the longitudes at any moment and compute the tithi yourself. Compare with a published *pañcāṅga*.
- **One tier down:** Memorise only the 6 anchor tithis — *pratipadā*, *pañcamī*, *aṣṭamī*, *ekādaśī*, *caturdaśī*, *pūrṇimā/amāvasyā*. Fill the rest in next week.

Teacher notes

- The 12°-per-tithi rule is the single most important computational fact in this module. Drill it.
- Watch for “tithi = day” confusion — every quiz will catch this. The simplest fix: ask “if a tithi started at 7 a.m. and the next started at 5 a.m. the next morning, how many hours was that tithi?” (22.) Repeat with different times until students see that tithi length varies.
- Pronunciation: *ṣaṣṭhī* is hard — “shash-tee.” *Ekādaśī* — “ay-KAH-da-shee.” Don’t rush.

Citation(s) used in this lesson

- Paraphrase from *Sūrya-siddhānta*, chapter 2: the tithi is defined as the period during which the Moon and Sun separate by 12° of arc. Cited by text name only; we do not quote a specific verse.

Day 4 — Build a Tithi Calendar

Module: Moon Phases & Tithi · **Band:** Middle · **Time:** 45 min

Learning objective

Each student (or pair) will build a tithi calendar for the current calendar month — labelling each date with its tithi name, pakṣa, and expected Moon shape.

Materials

- One printed page of a current-month *pañcāṅga* per pair (Drik Panchang, Mypanchang, or any standard source). Confirm the source and timezone before printing.
- A blank monthly calendar grid (printed handout — see activity-01)
- Coloured pencils or markers (two colours — one for *śukla*, one for *kṛṣṇa*)
- Notebooks

Vocabulary reused today

- *tithi*, *pakṣa*, *śukla pakṣa*, *kṛṣṇa pakṣa*, *amāvasyā*, *pūrṇimā* (Days 1–3)
- All 15 tithi names (Day 3)

Lesson flow

Warm-up (5 min)

- Daily prompt: “*Did you see the Moon last night? What tithi was it?*” — 2 students share.
- Hand back yesterday’s formative quizzes briefly (90 sec) — call out the most common miss; correct it once.
- “*Today the puzzle is concrete: what’s the tithi for every date this month?*”

Core (15 min)

1. **Show one sample row.** Project or photocopy a *pañcāṅga* entry for any one date. Walk through:
 - Date: e.g. “May 22”
 - Pakṣa: e.g. *kṛṣṇa*
 - Tithi at sunrise: e.g. *caturthī*
 - Tithi end-time: e.g. 14:37 (after which *pañcamī* begins)

“*So if you went out at 8 p.m. on May 22, the tithi would already be pañcamī, not caturthī. The tithi listed at the top of a calendar date is the one current at sunrise.*”
2. **The pattern of a month — anticipated.** Most months have:
 - A *śukla pratipadā* (the day after *amāvasyā*)
 - A *pūrṇimā* (in the middle)
 - A *kṛṣṇa pratipadā* (the day after *pūrṇimā*)
 - A *kṛṣṇa amāvasyā* (close to the end)
 - 28–31 dated tithis (because of *kṣaya* / *vṛddhi* — see Day 3)
3. **Anatomy of a tithi calendar — the activity sheet.** Walk through the blank grid:
 - One row per date
 - Columns: Date | Day of week | Pakṣa | Tithi name | Tithi end-time | Expected Moon shape
4. **One festival hook.** Find one festival this month from the *pañcāṅga* and announce it: “*On dvitīyā of śukla pakṣa this month, there’s a regional festival called X — let’s see if we agree with the date.*”

Activity (20 min) — Build the calendar

This **IS** Activity 01 — see `activities/activity-01-build-tithi-calendar.md` for the full set of instructions.

In pairs: 1. Receive one printed *pañcāṅga* page for the current month. 2. Fill in the blank calendar grid for every date of the month. 3. Colour code: *śukla pakṣa* in yellow, *kṛṣṇa pakṣa* in grey/blue. 4. Sketch a small Moon shape for each date (or every 3rd date if running short on time). 5. Circle any *amāvasyā* and *pūrṇimā*.

Walk around. The most common error: students copy the wrong column from the *pañcāṅga*. Have them double-check 2–3 entries with you.

Wrap-up (5 min)

- Pairs hold up their finished calendars briefly. The teacher takes a quick scan.
- **Exit question (1 min, oral):** “Which date this month is *pūrṇimā*?” Class answers in chorus.
- **Preview Day 5:** “Tomorrow we name the 12 lunar months and link them to festivals.”

Student-facing content

Today you built a tithi calendar.

A tithi calendar links each date in a calendar month to: - the *pakṣa* it falls in - the *tithi* current at sunrise - the time the tithi ends - the expected shape of the Moon

Two patterns to notice: - There are usually 1 *amāvasyā* and 1 *pūrṇimā* per calendar month — sometimes 0 or 2 depending on month length. - A few tithis will have unusual entries — a *kṣaya* (skipped) or *vṛddhi* (spanning two days). Note them; they are the system working as designed.

Homework

- Continue the Moon diary. Compare last night’s sketch with the calendar you built today — does the shape match the predicted shape for that tithi?
- For Day 5: be ready to name **one festival in your family** (or community) that you know is tied to a specific tithi.

Differentiation

- **One tier up:** Look at the *pañcāṅga* for the month BEFORE this one. Find a tithi that was *kṣaya* (skipped). Explain in writing what happened on that date — the previous tithi was running at sunrise, then the next tithi started after sunrise and ended before the next sunrise. So the “skipped” tithi never had its own sunrise.
- **One tier down:** Fill in the date column and the tithi-name column only. Skip the end-times and the Moon shapes for this lesson.

Teacher notes

- **Choose ONE *pañcāṅga* source for the entire class.** Different *pañcāṅgas* may use different sunrise locations or computational systems and give slightly different tithi names for borderline dates. Pick one (Drik Panchang is well-known; Mypanchang.com is also reliable) and stick to it for the whole module.
- Some calendars list two tithis on a single date (e.g. “Caturthī upto 14:37, then Pañcamī”). This is normal. Have students enter the tithi at sunrise, then note the transition.
- The colour-coding (yellow for *śukla*, blue/grey for *kṛṣṇa*) makes the calendar legible at a glance. Insist on it.
- Bring a backup: if printing fails, project the *pañcāṅga* for the whole class to read together. Less efficient but workable.

Citation(s) used in this lesson

- The class *pañcāṅga* source (named on the board): use this consistently. The data is computed using the same astronomical principles described in the *Sūrya-siddhānta* — paraphrased only; no verse cited.

Day 5 — Lunar Months and the Festival Calendar

Module: Moon Phases & Tithi · **Band:** Middle · **Time:** 45 min

Learning objective

Students will name the 12 lunar months in order, explain how each month is named (after the *nakṣatra* of its *pūrṇimā*), and link at least three festivals to their tithi-and-month positions.

Materials

- Board with a 12-row table ready
- A printed list of major festivals across the 12 lunar months (one per pair)
- Notebooks

Vocabulary introduced today

- māsa* (IAST: *māsa*; lit. “month”) — a lunar month, ~29.53 days.
- nakṣatra* (IAST: *nakṣatra*; lit. “lunar mansion”) — one of 27 segments of the ecliptic, each ~13°20’ wide. (Full coverage in a later module — today’s mention is just for naming.)
- adhika māsa* (IAST: *adhika-māsa*; lit. “extra month”) — an intercalary 13th month inserted approximately every 32.5 months.

Lesson flow

Warm-up (5 min)

- Daily prompt: “*Did you see the Moon last night? What tithi?*” — 2 students share.
- Quick recall using yesterday’s tithi calendar: “*What month did we build the calendar for?*” (May, June, etc.) “*In the Indian lunar calendar, what month is that?*” (Answer: it depends — let’s find out.)

Core (20 min)

- The 12 lunar months.** Write the list on the board:

#	Lunar month (IAST)	Roughly corresponds to	Named after
1	<i>Caitra</i>	March–April	nakṣatra <i>Citrā</i>
2	<i>Vaiśākha</i>	April–May	<i>Viśākhā</i>
3	<i>Jyeṣṭha</i>	May–June	<i>Jyeṣṭhā</i>

#	Lunar month (IAST)	Roughly corresponds to	Named after
4	<i>Āṣāḍha</i>	June–July	<i>Pūrva- / Uttara-āṣāḍhā</i>
5	<i>Śrāvaṇa</i>	July–August	<i>Śrāvaṇa</i>
6	<i>Bhādrapada</i>	August–September	<i>Pūrva- / Uttara-bhādrapadā</i>
7	<i>Āśvina</i>	September–October	<i>Āśvinī</i>
8	<i>Kārtika</i>	October–November	<i>Kṛttikā</i>
9	<i>Mārgaśīrṣa</i>	November–December	<i>Mṛgaśīrṣa</i>
10	<i>Pauṣa</i>	December–January	<i>Puṣya</i>
11	<i>Māgha</i>	January–February	<i>Maghā</i>
12	<i>Phālguna</i>	February–March	<i>Phālgunī</i> (Pūrva / Uttara)

2. **Why these names? The naming convention.** A lunar month is named after the *nakṣatra* in which the *pūrṇimā* (full moon) of that month falls. The Moon at full moon is opposite the Sun, so it sits in a specific star background — the *nakṣatra*. The lunar month inherits its name from that *nakṣatra*.

3. **Two month-counting conventions.** This is a real difference and worth noting:

- *Amānta* — the lunar month ends at *amāvasyā*. Common in South and Western India (Maharashtra, Karnataka, Andhra, Tamil Nadu, Gujarat).
- *Pūrṇimānta* — the lunar month ends at *pūrṇimā*. Common in North India (Uttar Pradesh, Bihar, North Indian Hindi belt).

So the same fortnight can be called by different month names under the two conventions! For example, the *kṛṣṇa pakṣa* before Janmāṣṭamī is “*Śrāvaṇa kṛṣṇa pakṣa*” under *pūrṇimānta* and “*Bhādrapada kṛṣṇa pakṣa*” under *amānta*. Both are correct — they are counting from different anchor points.

4. **Linking festivals to tithis — the festival table.** Write 6 examples:

Festival	Lunar month + pakṣa + tithi	What it celebrates / who observes
<i>Ugādi / Gudi Pādvā</i>	<i>Caitra śukla pratipadā</i>	Lunar new year (in much of South / West India)
<i>Rāma Navamī</i>	<i>Caitra śukla navamī</i>	Birth of Rāma
<i>Akṣaya Tṛtīyā</i>	<i>Vaiśākha śukla tṛtīyā</i>	Auspicious for beginnings

Festival	Lunar month + pakṣa + tithi	What it celebrates / who observes
<i>Guru Pūrṇimā</i>	<i>Āṣāḍha pūrṇimā</i>	Honouring teachers
<i>Janmāṣṭamī</i>	<i>Bhādrapada</i> (or <i>Śrāvaṇa</i> , depending) <i>kṛṣṇa aṣṭamī</i>	Birth of Kṛṣṇa
<i>Dīpāvalī</i> / <i>Dīwālī</i>	<i>Kārtika amāvasyā</i>	Festival of lights
<i>Mahā-śivarātri</i>	<i>Phālguna</i> (or <i>Māgha</i>) <i>kṛṣṇa caturdaśī</i>	Great night of Śiva
<i>Holī</i>	<i>Phālguna pūrṇimā</i>	Festival of colours

Every festival here has a precise tithi-and-month coordinate. That’s the whole point of the *pañcāṅga* system.

5. **The intercalation puzzle — *adhika māsa*.** 12 lunar months × 29.53 days = 354.36 days. The solar year is 365.24 days. So the lunar calendar drifts behind by ~11 days a year. After 3 years that’s 33 days — about a whole month. To re-align, the Indian system inserts an *adhika māsa* (“extra month”) roughly every 32.5 months. This is what makes the Indian calendar **lunisolar** rather than purely lunar. (More on this in the cross-cultural lesson, Day 9.)

Activity (15 min) — Festival mapping + formative quiz

Part A (8 min) — pair work. Each pair gets the printed festival list (about 12 festivals). Their job:

1. Mark which lunar month each festival falls in.
2. Identify the tithi and pakṣa.
3. Find ONE festival that no one in the pair had heard of before. Write its month + pakṣa + tithi.

Part B (7 min) — formative quiz, see quizzes/formative.md Part B. Collect and mark within 24 hours.

Wrap-up (5 min)

- One student names a festival their family celebrates; the class names the tithi+pakṣa+month. Two rounds.
- **Project decision deadline:** by the end of today, every student has a project partner (or has confirmed individual work) and a tentative topic. Sticky note on the board.

Student-facing content

There are 12 lunar months. Each is named after the *nakṣatra* in which its *pūrṇimā* falls.

Caitra, Vaiśākha, Jyeṣṭha, Āṣāḍha, Śrāvaṇa, Bhādrapada, Āśvina, Kārtika, Mārgaśīrṣa, Pauṣa, Māgha, Phālguna.

Two conventions: months end at *amāvasyā* (*amānta*) or at *pūrṇimā* (*pūrṇimānta*).
Different regions, different conventions — same astronomy.

Every Indian festival sits at a precise (month + pakṣa + tithi) coordinate. The *pañcāṅga* tells you exactly when.

Homework

- Continue the Moon diary.
- Submit your **project plan paragraph** — 5–8 sentences stating which lunar phenomenon you’ll cover (phases / eclipses / intercalary month / something else), in what format, and what your audience walks away with.

Differentiation

- **One tier up:** *Adhika māsa* deep-dive. Look up the rule: an *adhika māsa* is a lunar month in which no *saṁkrānti* (Sun’s entry into a new zodiac sign) occurs. Why does the rule produce ~7 extra months in 19 years?
- **One tier down:** Memorise only the first 6 month names this week. The rest can come during the project.

Teacher notes

- The *amānta* / *pūrṇimānta* distinction confuses everyone the first time. It’s OK to leave students slightly fuzzy on it — the key insight is that conventions vary regionally, and “the calendar” isn’t one fixed thing.
- Some students will know a festival under a regional name (e.g. *Gudi Pādvā* vs. *Ugādi* — both *Caitra śukla pratipadā*). Honour both.
- The “12 months are named after nakṣatras” point doesn’t need full nakṣatra coverage today — that comes in a later module. Today we just say *what* the naming convention is.

Citation(s) used in this lesson

- General reference: *pañcāṅga* sources (named in class). Paraphrase from *Sūrya-siddhānta*, chapter 14 (computation of *adhika māsa*) — cited by text and chapter name only.

Day 6 — The Pañcāṅga: Five Limbs

Module: Moon Phases & Tithi · **Band:** Middle · **Time:** 45 min

Learning objective

Students will name the five limbs (*aṅga*) of the *pañcāṅga* — tithi, vāra, nakṣatra, yoga, karaṇa — and explain in one sentence each what each limb tracks.

Materials

- The *pañcāṅga* sample page used on Day 4
- Board with a 5-row table ready
- Notebooks

Vocabulary introduced today

- *pañcāṅga* (IAST: pañca-aṅga; lit. “five limbs”) — the traditional Indian almanac listing five astronomical quantities for each day.
- *aṅga* (IAST: aṅga; lit. “limb, part”) — one of the five components.
- *vāra* (IAST: vāra; lit. “day of the week, occasion”) — the weekday.
- *yoga* (IAST: yoga; lit. “joining, combination”) — here, one of 27 astronomical divisions based on the sum of Sun and Moon longitudes. (Distinct from yoga as in Patañjali’s discipline — same word, different application.)
- *karaṇa* (IAST: karaṇa; lit. “doing, half-portion”) — a half-tithi. There are 60 karaṇas in a synodic month.

Lesson flow

Warm-up (5 min)

- Daily prompt: “*Did you see the Moon last night? What tithi?*” — 2 students share.
- “*So far, we’ve spent five days on ONE limb of the pañcāṅga — the tithi. The pañcāṅga has FOUR more limbs. Today: a tour.*”

Core (25 min)

1. **The five limbs.** Write on the board:

#	Limb	What it tracks	Number of options
1	<i>tithi</i>	Moon’s elongation from Sun	30 per month (covered Days 3–4)
2	<i>vāra</i>	day of the week	7
3	<i>nakṣatra</i>	Moon’s position against the star background	27
4	<i>yoga</i>	sum of Sun-and-Moon ecliptic longitudes	27
5	<i>karaṇa</i>	half-tithi	11 names cycling (because of how the 60 karaṇas of a month are named)

2. **Limb-by-limb walkthrough.** Spend 3 min on each, with concrete reference to the Day 4 *pañcāṅga* page:

Vāra — the weekday (7). Named after the seven classical bodies: Sun (*Ravi*), Moon (*Soma* / *Indu*), Mars (*Maṅgala*), Mercury (*Budha*), Jupiter (*Guru* / *Brhaspati*), Venus (*Śukra*), Saturn (*Śani*). English Sunday/Monday/... has the same Sun/Moon/Saturn pattern — same naming logic from the same classical-period astronomy.

Nakṣatra — lunar mansion (27). The Moon takes about 27.3 days to circle the sky and return to the same star background (the *sidereal* month). Indian astronomers divided the Moon’s path into 27 segments, each ~13°20’ wide, named after the star pattern in that segment (*Āśvinī*, *Bharaṇī*, *Kṛttikā*, *Rohiṇī*, ... *Revatī*). The *pañcāṅga* lists which nakṣatra the Moon is in each day.

Yoga — joining (27). Add the Sun’s ecliptic longitude to the Moon’s. Divide by $13^{\circ}20'$. The result is the yoga number (1–27). Each has a name (Viṣkambha, Prīti, Āyusmān, ...). It’s a derived quantity, useful for timing in traditional astrology.

Karaṇa — half-tithi (11 cycling). Each tithi is split into two halves. The first 7 karaṇas of the month are *cara* (“moving”) karaṇas — Bava, Bālava, Kaulava, Taitila, Garaja, Vaṇija, Viṣṭi. They cycle 8 times through the month. The last 4 are *sthira* (“fixed”) — Śakuni, Catuṣpada, Nāga, Kṛṣṭughna — and appear once each near *amāvasyā*. So $8 \times 7 + 4 = 60$ half-tithis per month.

3. **Why five limbs?** Each limb encodes a different astronomical fact:

- *tithi*: position of the Moon **relative to the Sun**.
- *nakṣatra*: position of the Moon **against the stars**.
- *yoga*: combined Sun + Moon position.
- *vāra*: solar day count.
- *karaṇa*: a finer division for sub-day timing.

Together they pinpoint the astronomical situation on any given day. The *pañcāṅga* is, in effect, a compact astronomical state vector.

4. **Modern parallel.** A modern astronomy almanac (e.g. an ephemeris like NASA’s Horizons system) lists planet positions, lunar phase, day of week, eclipse data, etc. The *pañcāṅga* is exactly this kind of document — older format, same purpose. The major difference: the *pañcāṅga* also carries cultural / ritual tags (auspicious times, fast days), which a modern ephemeris does not.

Activity (10 min) — Read the *pañcāṅga*

In pairs, students take yesterday’s *pañcāṅga* page and find:

1. Today’s date row.
2. The tithi (already familiar).
3. The vāra (compare with the English weekday — does it match?).
4. The nakṣatra. Try to pronounce it.
5. The yoga and karaṇa names (these are new — just read them, no need to compute).

Each pair writes one sentence: “Today, [date], the Moon is in [nakṣatra] nakṣatra, it is [tithi] of the [pakṣa] pakṣa, and the vāra is [day].”

Wrap-up (5 min)

- Read aloud 2–3 students’ sentences. The class reciprocates with the meanings (“Mars day,” “Saturn day,” etc.).
- **Preview Day 7:** “Tomorrow, the big question — why do eclipses happen on *amāvasyā* and *pūrṇimā*, but not every month?”

Student-facing content

A *pañcāṅga* has five limbs. Each tracks a different astronomical fact.

Limb	What it tracks
<i>tithi</i>	Moon's elongation from the Sun (1 of 30)
<i>vāra</i>	day of the week (1 of 7)
<i>nakṣatra</i>	Moon's position against the stars (1 of 27)
<i>yoga</i>	sum of Sun + Moon longitudes (1 of 27)
<i>karaṇa</i>	half-tithi (1 of 11 cycling names)

Knowing all five for a given moment is like having a snapshot of where the Sun and Moon are. A modern astronomy almanac does the same job in a different format.

Homework

- Continue the Moon diary.
- For Day 7: read NASA's published lunar phase page for the current year. Note any solar or lunar eclipses listed for this year — when, where visible. Bring 1 example.

Differentiation

- **One tier up:** Compute the yoga number for today. You need the Sun's ecliptic longitude and the Moon's ecliptic longitude (from any astronomy app). Add them, divide by $13^{\circ}20'$, take the integer part + 1 — that's the yoga number. Compare with what the *pañcāṅga* says.
- **One tier down:** Memorise three limbs only — *tithi*, *vāra*, *nakṣatra*. The yoga and karaṇa terms can stay as “things I've heard of” for now.

Teacher notes

- The *yoga* limb is the most confusing — partly because “yoga” also means the spiritual discipline. Note this explicitly. They are different applications of the same Sanskrit root *yuj* (“to join”).
- The *karaṇa* naming is genuinely complicated. Don't try to teach all 11 names today — just enough that students know there are half-tithis and they have names.
- This is a content-heavy day. If the class is fading, move the activity to a paired worksheet for tomorrow and use today's time for slow walkthrough.

Citation(s) used in this lesson

- General reference: the class *pañcāṅga* source. The five-limb framework dates to early Indian astronomical literature (e.g. *Pañcasiddhāntikā* of Varāhamihira, 6th c. CE), but the modern *pañcāṅga* form is later. Paraphrased; no specific verse cited.

Day 7 — Eclipses: Why and When

Module: Moon Phases & Tithi · **Band:** Middle · **Time:** 45 min

Learning objective

Students will explain (with a diagram) why solar eclipses occur on *amāvasyā* and lunar eclipses on *pūrṇimā*, why eclipses do not happen every month, and what the safe ways to view a solar eclipse are.

Materials

- A small globe (or a large white ball — the Earth)
- A smaller dark ball (a tennis ball, painted dark — the Moon)
- A torch (the Sun)
- Two metres of string (to mark orbits)
- The activity-02 handout for the diagramming exercise
- Notebooks

Vocabulary introduced today

- *eclipse* — when one body's shadow falls on another (lunar eclipse: Earth's shadow on Moon) or one body passes between two others (solar eclipse: Moon between Sun and Earth).
- *ecliptic plane* — the plane of Earth's orbit around the Sun.
- *nodes* — the two points where the Moon's orbit crosses the ecliptic plane.

Lesson flow

Warm-up (5 min)

- Daily prompt: “*Did you see the Moon last night? What tithi?*” — 2 students share.
- “*You have a Moon diary now. Has any night's Moon been completely covered by Earth's shadow? Probably not. Why not? That's today.*”

Core (25 min)

Safety first (2 min, before anything else). Write on the board in capitals:

NEVER LOOK DIRECTLY AT A SOLAR ECLIPSE. Ordinary sunglasses do not protect your eyes. You need ISO 12312-2 certified eclipse glasses, or use pinhole projection.

Every student copies this into their workbook. We will revisit at end of class.

1. The basic geometry — three bodies in a line.

Set up the demo: torch (Sun) on one side, white ball (Earth) in middle, dark ball (Moon) on the other side. Show the two configurations:

- **Solar eclipse:** Sun — Moon — Earth (in that order). Moon casts a shadow on Earth. This can ONLY happen on *amāvasyā* (because that's the only time the Moon is between Sun and Earth).

- **Lunar eclipse:** Sun — Earth — Moon. Earth casts a shadow on Moon. This can ONLY happen on *pūrṇimā* (because that's the only time the Moon is opposite the Sun).

2. So why doesn't this happen every month?

*“Every month has an *amāvasyā* and a *pūrṇimā*. If geometry were the only thing, we'd have a solar eclipse every new moon and a lunar eclipse every full moon. Clearly we don't.”*

The reason: **the Moon's orbit is tilted by $\sim 5^\circ$ relative to Earth's orbit around the Sun.** Show this with the string and the two balls — Earth's orbit is one flat ring; the Moon's orbit is a smaller ring tilted by 5° .

Most months, when the Moon is at *amāvasyā* (between Earth and Sun), it's either above or below the Sun-Earth line — its shadow misses Earth. Same for *pūrṇimā* — Earth's shadow misses the Moon.

An eclipse only happens when the Moon is at *amāvasyā* OR *pūrṇimā* AND it's near one of the two nodes (the points where the Moon's tilted orbit crosses the flat ecliptic plane). That's why eclipses come in pairs (a solar and a lunar within ~ 14 days) and in “eclipse seasons” twice a year.

3. How many eclipses per year?

- Solar: 2 to 5 per year, somewhere on Earth.
- Lunar: 0 to 3 per year.
- From any one location, a total solar eclipse is rare — the path of totality is narrow, and most observers see only partial eclipses.

4. Solar vs. lunar — what they look like.

- **Solar eclipse:** the Sun briefly turns into a “ring of fire” (annular) or vanishes completely (total). Daylight goes blue-grey. Birds may roost. Only visible from the narrow shadow strip.
- **Lunar eclipse:** the Moon turns dim coppery-red (“blood moon”) because the only sunlight reaching it is sunlight bent through Earth's atmosphere. Visible from the entire night-side of Earth. Lasts up to ~ 100 minutes.

5. Indian astronomical sources knew this.

The *Sūrya-siddhānta* (c. 5th–10th c. CE) gives algorithms for eclipse prediction — start time, duration, magnitude. Āryabhaṭa (5th c. CE) had a clear understanding of eclipse geometry (Moon's shadow, Earth's shadow, ecliptic plane) and rejected mythological explanations (the “Rāhu swallowing the Sun” story is older and pre-scientific; classical Indian astronomers had a separate, mathematical account).

Activity (10 min) — Diagram the geometry

This is `activities/activity-02-eclipse-geometry.md` in full. Students:

1. Receive the activity sheet (with two unfinished diagrams — Sun in one position, Earth in another, Moon’s orbit drawn around Earth).
2. Place the Moon (a sticker or a drawn circle) at the eclipse positions: between Sun and Earth (solar) and opposite (lunar).
3. Draw the shadow cones.
4. Mark the two nodes on the Moon’s orbit and shade the regions where an eclipse is possible.
5. Below the diagram, in 3 sentences, explain: *why does an eclipse not happen every amāvasyā / pūrṇimā?*

Wrap-up (5 min)

- Read the safety rule aloud as a class. Once more.
- Ask: “*What’s the safest way to watch a solar eclipse?*” (Eclipse glasses with ISO certification; or pinhole projection.) “*What about a lunar eclipse?*” (Look directly — no danger.)
- **Preview Day 8:** “*Tomorrow you start your project.*”

Student-facing content

Solar eclipse: Sun, Moon, Earth in line (Moon in middle). Happens on *amāvasyā*. Moon’s shadow falls on Earth.

Lunar eclipse: Sun, Earth, Moon in line (Earth in middle). Happens on *pūrṇimā*. Earth’s shadow falls on Moon.

Why not every month? The Moon’s orbit is tilted $\sim 5^\circ$ from Earth’s orbit around the Sun. So usually the Moon’s shadow misses Earth (at *amāvasyā*) or Earth’s shadow misses the Moon (at *pūrṇimā*). Eclipses happen only when the Moon is *both* at *amāvasyā/pūrṇimā* *and* close to one of the two nodes (where the two orbital planes cross).

⚠️ **NEVER look directly at a solar eclipse.** Use ISO 12312-2 certified eclipse glasses, or project the Sun through a pinhole onto paper. Sunglasses, smoked glass, photographic film do not work. Lunar eclipses are safe to view directly.

Homework

- Continue the Moon diary.
- Look up: what is the date of the next solar eclipse and the next lunar eclipse visible from India? Note in your workbook.

Differentiation

- **One tier up:** Find the dates of the nodes for this year. (The two nodes drift backward through the ecliptic — they complete one full circuit in ~ 18.6 years. This is the *Saros cycle*’s underlying mechanism.) Why does this give us “eclipse seasons” twice a year?
- **One tier down:** Skip the node-drift detail. Focus on: solar eclipse = *amāvasyā* + alignment, lunar eclipse = *pūrṇimā* + alignment. The “tilted orbit” is the key idea; the nodes are the precise version of it.

Teacher notes

- This is the single highest-safety-risk day in the entire IKS curriculum if there's a solar eclipse during the school term. Even if there isn't one this term, drill the safety rule. Eclipses come around.
- Some students will know the *Rāhu* story (Rāhu the demon swallowing the Sun/Moon during eclipses). Acknowledge it as a pre-scientific narrative shared by many cultures, then return to the geometric explanation. Classical Indian astronomers like Āryabhaṭa explicitly rejected the *Rāhu* explanation in favour of the shadow-geometry account.
- The “5° tilt” number is the key. Repeat it. Without it, the “why not every month” question has no answer.

Citation(s) used in this lesson

- Paraphrase from *Sūrya-siddhānta*, chapters 4–5 (computation of solar and lunar eclipses): the text gives detailed algorithms for predicting eclipse timing using the geometry of the Moon's nodes. Cited by text and chapter range only.
- Modern reference: NASA's eclipse pages (search “NASA Eclipse Website” — official, free).

Day 8 — Project Session 1: Plan and First Artefact

Module: Moon Phases & Tithi · **Band:** Middle · **Time:** 45 min

Learning objective

Students will produce a one-paragraph project plan with element (lunar topic), format, audience, and citation; and complete a first-draft artefact (poster sketch / video storyboard / slide outline).

Materials

- Project supplies for whatever format students chose: poster paper, markers, A4 sheets, laptops, art supplies
- Printed copy of the rubric (`assessments/rubric.md`) — one per pair
- The class *pañcāṅga* (for reference)
- Days 1–7 student notebooks (their own diaries + worksheets)

Lesson flow

Warm-up (5 min)

- Daily prompt: “Did you see the Moon last night? What tithi?” — 2 students share.
- Hand each pair a printed rubric. “Read through the rubric — 4 minutes. Bring questions to me.”

Core (5 min)

Take questions on the rubric. The most common questions:

- “Do we need to quote a verse?” — No. A paraphrase of the *Sūrya-siddhānta* (e.g. “the *Sūrya-siddhānta* defines a tithi as 12° of elongation”) is enough. A NASA reference is also acceptable. The rubric cares that your source is real and named.
- “Can we do TWO topics?” — One main topic. You may compare to a second briefly.
- “What if our diary is missing some nights (clouds)?” — Note “cloudy, predicted [shape]” for those nights. The diary still counts.

Activity (30 min) — Build

Pairs work on their projects. The teacher rotates and checks each pair has:

1. **Chosen topic** — one of:
 - Moon phases (geometry and the 10-night diary visualised)
 - Eclipses (why, when, safety)
 - Tithi calendar (the one built on Day 4 + an explanation)
 - Cross-cultural lunar calendars (preview of Day 9)
 - Lunar months and festivals
 - Adhika māsa (intercalary month)
2. **Chosen format** — one of:
 - Poster (A2 or larger)
 - Slide deck (5–8 slides, projected)
 - 90-second video
 - 1-page printed story / explainer
 - Live demonstration with the torch and ball
 - Infographic / data visualisation
3. **One source named** — *Sūrya-siddhānta* (paraphrased), NCERT Physics XI, NASA’s phase calendar, the class *pañcāṅga*, etc. **Real sources only.**
4. **By end of class:** a one-paragraph plan + a first-draft artefact (a poster sketch, video storyboard, slide outline).

The teacher intervenes if: - A pair is still debating topic at minute 15 — give them 5 minutes, then assign. - A pair’s source is invented or unverifiable — flag it; help them swap. - A pair has split unequally — re-balance the workload before they leave.

Wrap-up (5 min)

- Each pair submits their plan paragraph + first-draft artefact (phone photo accepted).
- **Tomorrow:** finish the build. The cross-cultural calendar lesson runs in parallel — 15 min content, then 30 min build time. Be ready.

Student-facing content

Today is build day.

By the end of this class, your pair (or you, if solo) should have: - A one-paragraph plan in writing. - A first-draft artefact — poster sketch, slide outline, video storyboard. - Your source named, on the artefact.

Tomorrow you refine. Day 10 you present.

Homework

- Continue the Moon diary (you have 2 nights left).
- Finish at home what you couldn't finish in class. Don't *start* something new — refine.

Differentiation

- **One tier up:** Add a second citation. Two sources is a stronger project — for instance, paraphrase the *Sūrya-siddhānta* AND cite NASA's phase calendar.
- **One tier down:** Reduce scope. Pick ONE phenomenon (phases, eclipses, or the calendar) and explain it in one paragraph + one diagram. That's a complete project.

Teacher notes

- Pairs that are still “researching” by minute 20 are stuck. Push them to start sketching/drafting even if incomplete. The artefact concretises the thinking.
- Phone-photo submissions are fine. Photograph the draft poster against a wall; submit by email or via shared drive.
- A common failure mode: pairs over-design the visual and under-write the explanation. Push toward writing the **3-minute script** by end of class.

Citation(s) used in this lesson

- The rubric (`assessments/rubric.md`) is the primary reference for what “good” looks like.
- Source-quality reminder: ONLY real sources. No invented citations.

Day 9 — Cross-cultural Lunar Calendars + Project Build

Module: Moon Phases & Tithi · **Band:** Middle · **Time:** 45 min (15 min content + 30 min project)

Learning objective

Students will compare the Indian lunisolar calendar with three other lunar/lunisolar calendars (Islamic Hijri, Chinese, Hebrew) — identifying what is shared (lunar month basis) and what differs (intercalation rules).

Materials

- Board with a comparison table ready
- Project supplies (continued from Day 8)
- Notebooks

Lesson flow

Warm-up (5 min)

- Daily prompt: “*Did you see the Moon last night? What tithi?*” — 2 students share.
- “*Many cultures, looking at the same Moon, built different calendars. Today: a 15-minute compare-and-contrast, then back to project build.*”

Core content (15 min)

1. **The common starting point.** Every culture observing the Moon notices the ~29.5-day cycle. The question every culture answers: *how do we fit 29.5-day lunar months into a 365.24-day solar year?*

The three resolutions:

- **Ignore the solar year.** Use only lunar months. The months drift through the seasons. (Islamic / Hijri)
 - **Re-align periodically with an extra month.** Lunisolar. (Indian, Chinese, Hebrew)
 - **Ignore the lunar month.** Use only the solar year, fix months at 30–31 days. (Modern Gregorian)
2. **The four calendar table.** Build on the board:

Feature	Indian (Hindu)	Islamic (Hijri)	Chinese	Hebrew
Type	Lunisolar	Pure lunar	Lunisolar	Lunisolar
Months/ year	12 (or 13 with <i>adhika māsa</i>)	12	12 (or 13 with leap month)	12 (or 13 with <i>Adar II</i>)
New month begins at	<i>amāvasyā</i> or <i>pūrṇimā</i> (region-dependent)	First sighting of crescent after new moon	<i>amāvasyā</i> (new moon)	<i>amāvasyā</i> (with calculated rules)
Days per year (average)	~365.25	~354	~365.25	~365.25
Intercalati on rule	Month with no <i>saṁkrānti</i> becomes <i>adhika</i> (extra)	None	Month with no <i>zhōngqì</i> (zhongqi solar term)	7 leap months in 19 years (Metonic)
Drift through seasons	Stays aligned	Drifts ~11 days/year backward	Stays aligned	Stays aligned

3. **A specific example.** *Ramaḍān* (Islamic) shifts ~11 days earlier each Gregorian year. *Caitra śukla pratipadā* (Indian new year, *Ugādi/Gudi Pādvā*) stays in March–April. *Pesach* (Hebrew) stays in spring. *Chūn Jié* (Chinese new year) stays in late January /

early February. The lunisolar systems all use intercalation to keep their months in-season; the pure lunar system does not.

4. **Why does any of this matter?** Three reasons:

- **Cultural literacy.** Knowing how other calendars work helps in a multi-cultural society.
- **Astronomical insight.** Each calendar reveals a different choice in handling the same underlying mismatch (29.5 vs. 365.24).
- **Historical depth.** All four systems were refined over many centuries. The Indian, Chinese, and Hebrew systems independently arrived at lunisolar intercalation — convergent astronomical reasoning.

Activity (25 min) — Project build

Pairs continue their projects from Day 8. The teacher rotates with two priorities:

1. **Source check.** Every project has a real, named citation by end of class today. No invented references.
2. **3-minute script check.** Every pair has written or rehearsed a 3-minute presentation script. The artefact alone is not the project; the *explanation* is.

Pairs whose project relates to cross-cultural lunar calendars can use the Day 9 content directly. Others are encouraged to add a one-sentence cross-cultural note (“this Indian system, like the Chinese and Hebrew, is lunisolar…”).

Wrap-up (5 min)

- All pairs submit a near-final draft (photo / file).
- One sentence from each pair on the board: **“Our project covers ____ and our source is ____.”**
- **Tomorrow:** Day 10. 3-minute presentations. Summative quiz.

Student-facing content

Four lunar/lunisolar calendars compared:

- **Indian (lunisolar)** — 12 months, *adhika māsa* every ~32 months.
- **Islamic / Hijri (pure lunar)** — 12 months, no intercalation. Drifts ~11 days/year.
- **Chinese (lunisolar)** — 12 months, leap month every ~32 months.
- **Hebrew (lunisolar)** — 7 leap months in 19 years (Metonic cycle).

The Indian, Chinese, and Hebrew systems independently arrived at lunisolar intercalation. Their rules differ; the underlying problem is the same.

Homework

- Continue the Moon diary (one night left).
- Final touches on your project. Bring everything to class tomorrow.

Differentiation

- **One tier up:** Pick TWO calendars and write a paragraph: under what circumstances does each calendar's design choice make sense? (Pure lunar — when religious months not tied to season matter most; lunisolar — when agriculture matters; pure solar — when civil administration matters.)
- **One tier down:** Compare just the Indian and the Hijri (pure lunar). One difference, one similarity. That's enough.

Teacher notes

- This content is genuinely cross-cultural. Avoid framing as “ours is best.” All four systems work; they encode different priorities.
- The Metonic cycle (19-year Hebrew cycle) and the Indian intercalation rule converge to about the same accuracy. This is mathematically interesting — students at Senior band can explore it.
- Keep this part to 15 minutes. The project is the major work today.

Citation(s) used in this lesson

- General references for each calendar — use any standard encyclopaedic source (e.g. *Encyclopaedia Britannica* entries on the Hijri, Chinese, and Hebrew calendars). Do not invent specific verse numbers from primary sources.

Day 10 — Final Assessment + Showcase

Module: Moon Phases & Tithi · **Band:** Middle · **Time:** 45 min (split: 20 min quiz + 25 min showcase)

Learning objective

Students demonstrate the module's full set of objectives through a summative quiz, and present their projects in 3 minutes per pair to the class.

Materials

- Printed summative quiz (`quizzes/summative.md`) — one per student
- All projects ready to present
- A visible timer (phone, stopwatch)
- The rubric (`assessments/rubric.md`) for marking

Lesson flow

Warm-up (3 min)

- Daily prompt: “*Did you see the Moon last night? What tithi?*” — 2 students share. (Last time for the diary prompt.)
- “*Two things today — quiz first, then the showcase. Pencils ready.*”

Summative quiz (20 min)

- Distribute the quiz.
- 20 minutes, closed book, individual.
- Strict silence after the start signal.
- Collect at end. Mark over the weekend.

Showcase (20 min)

- Pairs present in turn — 3 minutes each + 1 minute Q&A.
- Visible timer running. Strict cutoff at 3:30.
- Teacher marks on the rubric in real time.
- Audience: students take notes — what's one fact they didn't know, from each pair?

(For a class of 30 students = 15 pairs, this won't fit in 20 minutes. **Either split presentations across Day 9 and Day 10, or schedule a separate showcase session.** See the lesson-plan note.)

Wrap-up (2 min)

- *“You’ve each kept a 10-night Moon diary. You’ve built a tithi calendar. You’ve explained eclipses. You’ve presented a project. Look up tonight — you’ll see the Moon a little differently.”*
- Reflection slips submitted with projects.
- Collect Moon diaries.

Student-facing content

Module complete.

Today: 1. Summative quiz — 20 min, individual, closed book. 2. Showcase — 3 minutes per pair + 1-minute Q&A. 3. Reflection slip submitted.

By end of class, you’ve shown what you learned. Take a breath.

Homework

- **None.** Module complete.

Differentiation

- **One tier up:** Volunteer to be the audience-facing rubric scorer (anonymously) — practice the discipline of assessing peers fairly.
- **One tier down:** Students who couldn't finish their project today have until next class to submit. Don't penalise on Day 10 if the artefact is 80% done.

Teacher notes

- Mark the quiz over the weekend; return within 3 school days.
- Project rubric marks should be visible to students within 5 school days.
- Reflection slips are NOT graded but are read — write one-sentence response on each.
- Keep one or two strong projects (with student permission) as exemplars for next year.

Citation(s) used in this lesson

- N/A — this is assessment day.

Module 6 (Middle) — Formative Quizzes

Two parts: Part A on Day 3 (after tithi naming), Part B on Day 5 (mid-module). Both are diagnostic — not graded. Hand back within 24 hours.

Part A — Day 3 (3 min, 5 questions)

Name: _____ Class: _____

1. A *tithi* is defined as the time during which the Moon gains _____ degrees of elongation on the Sun.

- a) 12
- b) 15
- c) 24
- d) 30

2. How many tithis are there in one synodic month?

- a) 15
- b) 27
- c) 28
- d) 30

3. True or false: A tithi always lasts exactly 24 hours. (T / F)

4. Name the tithi that corresponds to the full moon.

5. Translate: *aṣṭamī*.

Part A — answer key

Q	Answer	Notes
1	a) 12	$360^\circ / 30 = 12^\circ$.
2	d) 30	15 per pakṣa × 2 pakṣas.
3	F	Average ~23h 37m; can range ~21.5–26h.

Q	Answer	Notes
4	<i>pūrṇimā</i>	Spelling tolerated if recognisable.
5	Eighth	Sanskrit ordinal for 8.

Part B — Day 5 (10 min, 5 questions, mixed format)

Name: _____ Class: _____

1. (2 marks) Write the names of the two pakṣas, and what happens to the Moon during each.

2. (3 marks) Complete the table:

Anchor day	Sanskrit name	Tithi number in pakṣa
New moon	_____	_____
Full moon	_____	_____
First day after new moon	_____	1

3. (2 marks) Match each lunar month to its season (draw lines).

Caitra	Late autumn (Oct–Nov)
Śrāvaṇa	Spring (Mar–Apr)
Kārtika	Summer monsoon (Jul–Aug)
Mārgaśīrṣa	Early winter (Nov–Dec)

4. (1 mark) Why is the Indian calendar called “lunisolar”?

5. (2 marks, short answer) Why can a tithi sometimes “skip a sunrise” (*kṣaya*)?

Part B — answer key

Q	Answer	Mark scheme
1	<i>śukla pakṣa</i> — Moon grows (waxes), from new to full; <i>kṛṣṇa pakṣa</i> — Moon shrinks	2 if both names + both directions; 1 if either part incomplete.

Q	Answer	Mark scheme
	(waning), from full to new.	
2	New moon = <i>amāvasyā</i> , 15 (of <i>kṛṣṇa pakṣa</i>). Full moon = <i>pūrṇimā</i> , 15 (of <i>śukla pakṣa</i>). First-day-after = <i>pratipadā</i> .	3 (1 per row); 0.5 for partial.
3	Caitra–Spring; Śrāvaṇa–Monsoon; Kārtika–Late autumn; Mārgaśīrṣa–Early winter.	2 if all 4; 1 if 2–3.
4	Because it uses lunar months for naming (29.5-day months) but re-aligns periodically with the solar year using an <i>adhika māsa</i> .	1 if both lunar and solar components named.
5	A tithi has an average length of ~23h 37m. The Moon doesn’t move uniformly. So sometimes a tithi begins after one sunrise and ends before the next — its short duration means it never overlaps with a sunrise.	2 if both “tithi length varies” AND “shorter than a day” reasoning appear. 1 if either part is present.

Total Part B: 10 marks. Not added to summative.

What to do with the results

- A class average below 6/10 on Part B = significant gaps. Reteach the pakṣa names and the tithi-vs-day distinction on Day 6 (before moving to the *pañcāṅga* tour).
- Individual students below 3/10 on Part B = pull aside on Day 7 for a 10-min one-on-one before group activity starts.

Module 6 (Middle) — Summative Quiz

Time: 20 min · **Marks:** 25 · **Closed book**

Instructions: Write your name and class. Answer all questions. Section D has one mark for clear handwriting and sentence structure.

Name: _____ **Class:** _____

Section A — Multiple choice ($5 \times 1 = 5$ marks)

1. A *tithi* is defined as 1/30 of a:

- a) calendar day
- b) synodic month
- c) sidereal month
- d) solar year

2. The waning fortnight is called:

- a) *śukla pakṣa*
- b) *pūrṇimā pakṣa*
- c) *kṛṣṇa pakṣa*
- d) *amāvasyā pakṣa*

3. A solar eclipse can occur only on:

- a) *pūrṇimā*
- b) *aṣṭamī*
- c) *ekādaśī*
- d) *amāvasyā*

4. Which limb of the *pañcāṅga* tracks the Moon's position against the star background?

- a) *tithi*
- b) *vāra*
- c) *nakṣatra*
- d) *karaṇa*

5. True/false (circle one): A *tithi* always lasts 24 hours. T / F

Section B — Short answer ($4 \times 2 = 8$ marks)

6. Why does the Moon appear to change shape over the month? Use 2–3 sentences.

7. Name the five limbs of the *pañcāṅga*.

8. Why don't eclipses happen every *amāvasyā* and every *pūrṇimā*?

9. What is *adhika māsa* and why does the Indian calendar need it?

Section C — Diagram + working ($1 \times 5 = 5$ marks)

10. Draw the Sun–Earth–Moon geometry for **both** a solar eclipse and a lunar eclipse. Label each diagram with: (a) which body is between the others, (b) which tithi the eclipse occurs on, (c) which body’s shadow is involved.

[draw solar eclipse here]

[draw lunar eclipse here]

Section D — Open response ($1 \times 6 = 6$ marks; 1 of which is for clarity)

11. Pick ONE feature of the Indian *pañcāṅga* system. In 4–5 sentences, describe: - what astronomical fact it captures - one practical or cultural use of it - one specific question you still have about it

Answer key (teacher copy)

Q	Answer	Mark scheme
1	b) synodic month	1
2	c) <i>kṛṣṇa pakṣa</i>	1
3	d) <i>amāvasyā</i>	1
4	c) nakṣatra	1
5	F	1
6	The Sun lights up only half of the Moon. As the Moon orbits Earth, different fractions of the lit half face Earth. We see the changing fractions as phases.	2 if all three of (Sun lights half) + (orbital motion) + (different fractions visible) are present. 1 if two of the three.
7	<i>tithi, vāra, nakṣatra, yoga, karaṇa</i>	2 if all 5; 1 if 3–4; 0 if fewer. Spelling tolerated.
8	The Moon’s orbit is tilted by ~5° relative to Earth’s orbit around the Sun. Most months, the Moon’s shadow misses Earth (or Earth’s shadow misses the Moon) at the alignment. Eclipses happen only when the Moon is at one of its two orbital nodes near <i>amāvasyā/pūrṇimā</i> .	2 if the tilted-orbit AND node ideas both appear; 1 if only one.
9	<i>Adhika māsa</i> is an intercalary “extra month” inserted approximately every 32.5 months in the Indian calendar. Without it, the lunar months ($12 \times 29.5 = 354$ days) would drift 11 days behind the solar year (365.24) each year, and festivals would slowly leave their season.	2 if both “intercalary month” AND “to re-align lunar with solar” are present. 1 if either alone.
10	Two diagrams: (a) Solar: Sun → Moon → Earth, on <i>amāvasyā</i> , Moon’s shadow on Earth. (b) Lunar: Sun → Earth → Moon, on <i>pūrṇimā</i> , Earth’s shadow on Moon.	5: 1 for each of (correct order) × 2; 1 for each tithi label × 2; 1 for clarity.
11	A specific feature (<i>tithi</i> , <i>pakṣa</i> , <i>nakṣatra</i> , <i>yoga</i> , <i>karaṇa</i> ,	6: 2 for astronomical accuracy; 2 for practical use; 1 for genuine question; 1 for clarity of writing.

Q	Answer	Mark scheme
	adhika māsa, eclipse prediction, etc.) with astronomical fact, practical use, and a real question.	

Total: 25 marks.

Grading bands (for project combined): - 32–40: Exceeds - 24–31: Meets - 16–23: Approaches - 0–15: Beginning

Homework

Homework — Module 6 Moon Phases & Tithi (Middle)

Homework tasks per day. None should take more than 25 minutes for a Middle-band student. Skip a day if it conflicts with another subject’s deadline.

The **Moon diary** is a running task: every night from Day 1 to Day 10, sketch the Moon’s shape, note the date/time/cloud cover, and (from Day 3 onwards) name the tithi.

Day 1 → Day 2

Task: Start your **10-night Moon diary**. Tonight: look at the Moon. Sketch its shape in your workbook. Note the date, time, and “clear / partly cloudy / overcast.” If overcast, look up the phase on a Moon-phase app or NASA’s published phase calendar.

Time: 5 min (looking) + 5 min (sketching).

Day 2 → Day 3

Task: Continue the Moon diary. Tonight, add **which pakṣa you are in** to your sketch.

Bonus: Find out from a parent or a calendar — has anyone in your family observed *amāvasyā* or *pūrṇimā* this month for any reason (fast, ritual, prayer)?

Time: 10 min.

Day 3 → Day 4

Task: Memorise the 15 tithi names (śukla pakṣa). Quiz yourself or be quizzed at home. **Be ready to recite all 15 in under 60 seconds at the start of tomorrow’s class.**

Continue the Moon diary — now naming the tithi as well as the pakṣa.

Time: 15 min.

Day 4 → Day 5

Task: Compare last night’s Moon-diary sketch with the tithi calendar you built in class. Does the shape match the predicted shape for the tithi?

Also: be ready to name **one festival in your family** (or community) that you know is tied to a specific tithi.

Time: 10 min.

Day 5 → Day 6

Task: Submit your **project plan paragraph** in writing.

Format: one paragraph, 5–8 sentences, stating: which lunar phenomenon, which format (poster / demo / story / video / infographic / slide deck), and what audience will see/hear/feel.

Time: 15 min.

Day 6 → Day 7

Task: Read NASA’s published lunar phase page for the current year (search “NASA Moon phases [year]”). Note any solar or lunar eclipses listed for this year — when, where visible. Bring 1 example.

Continue the Moon diary.

Time: 20 min.

Day 7 → Day 8

Task: Bring ALL project materials to tomorrow’s class. The build session starts immediately. Anything you forget cannot be retrieved during class time.

Bonus: Look up the date of the next solar eclipse and the next lunar eclipse visible from your location. Note in your workbook.

Time: 15 min.

Day 8 → Day 9

Task: Finish at home what you couldn't finish in class today. Tomorrow includes the cross-cultural lesson + refining, not starting.

Time: variable — but cap at 30 min, then stop and rest. Done-is-better-than-perfect.

Day 9 → Day 10

Task: Final touches. Make sure you have: - [] The artefact (poster / video file / printed story / etc.) - [] Your script or notes - [] Your completed Moon diary (10 nights) - [] A working version of your 3-minute presentation in your head

Time: 15 min.

After Day 10

No homework. Module is complete. Take a breath. (If your next module begins immediately, check that module's README for prerequisite reading.)

Optional follow-up: Keep the Moon diary going for another 20 days. By the end you'll have observed (or noted from a calendar) a full month of phases — and you'll know exactly which night to look for the next *pūrṇimā*.

Writing Prompts

Writing Prompts — Module 6 Moon Phases & Tithi (Middle)

Eight prompts in mixed modes. Assign as in-class quick-writes (10 min), homework (one paragraph), or as journal entries woven across the module. Each prompt names the lesson day where it fits naturally.

Reflection prompts

- 1. (Day 1, 5 min)** What did you think made the Moon change shape, before today? In 3–5 sentences, describe what you used to believe, and what changed your mind in class today. (If you already knew the geometry, write about a younger sibling or friend's belief instead.)
- 2. (Day 7, 10 min)** Stand outside for two minutes one evening (or open the window). Look up. Then list the Moon's position, its shape, and the tithi. In 4 sentences, describe whether the actual sky matched the *pañcāṅga* prediction. (If it didn't, what might explain the difference?)

Explanatory prompts (in your own words)

3. (Day 2 or 3, homework, ~150 words) Explain to a younger sibling (Grade 4) why a tithi is not the same as a day. Don't use the word "elongation" — find your own way to say it.

4. (Day 6, in-class, 10 min) Imagine you are explaining the *pañcāṅga* to a friend from another country who has never seen one. In 5 sentences, describe what the five limbs are and why having all five gives you a more complete picture than any one limb alone.

Argumentative prompt

5. (Day 6 or Day 7, homework, 1 paragraph) The modern Gregorian calendar ignores the Moon entirely — months are 30 or 31 days regardless of the Moon's actual phase. The Indian *pañcāṅga* tracks the Moon precisely but needs an *adhika māsa* every ~32 months to stay aligned with the seasons. Pick a side and defend it in 5–7 sentences: which approach is better for *modern civic life*, and which for *cultural/agricultural life*? Acknowledge the other side's strongest point before disagreeing.

Creative prompt

6. (Day 5 or Day 8, can be project — 1 page) Write a short scene (50–150 words) in which the Sun, Earth, and Moon have a conversation about why an eclipse is happening today. Each character gets at least one line. End with one of them explaining why this doesn't happen every month.

Cross-disciplinary prompt

7. (Day 6 or homework, ~120 words) The Indian *pañcāṅga* and a modern astronomy ephemeris (e.g. NASA Horizons) do similar work — both list astronomical facts for each day. Pick another field — geography, music, finance, sports — and describe a similar pairing: one traditional system and one modern system that do the same job in different formats. (Example: traditional Indian musicology's *rāga* system and modern Western music theory both organise pitch — different formats, same domain.)

Stretch prompt

8. (Optional, advanced students, Day 10 onwards, ~250 words) The *Sūrya-siddhānta* (Indian, c. 5th–10th c. CE) and the *Almagest* of Ptolemy (Greek, 2nd c. CE) both describe lunar motion in detail and both use geometric models with epicycles. They were developed independently. In a single page, argue for OR against the claim: "*Two cultures arriving at similar models of the same sky is evidence that the sky's behaviour, not the cultures' biases, drives astronomy.*" Use one specific example to support your argument.

Prompt-to-band notes

- All prompts are usable at Middle band as written.
- For **Primary** band, prompts 1, 2, 3, and 6 can be adapted (shorter, drawing-friendly).

- For **Senior** band, prompts 4, 5, 7, and 8 are essay-quality without modification; add a citation requirement.

How to give feedback

- Short prompts (1, 2): respond with one sentence of acknowledgement, not corrections. The point is observation.
- Explanatory (3, 4): respond with “the part that landed” and “the part still unclear.” Don’t correct grammar unless severely impeding meaning.
- Argumentative (5, 8): use the rubric — does the writing acknowledge the other side, give a specific example, and reach a clear position?
- Creative (6): no grading. Read aloud the best ones if you have permission.

Activity 01 — Build a Tithi Calendar

Module: Moon Phases & Tithi · **Band:** Middle · **Day:** 4 · **Time:** 20 min (within Day 4’s 45 min)

Type

Hands-on calendar build, paired

Goal

Each pair produces a tithi calendar for the current calendar month with every date labelled with its pakṣa, tithi, tithi end-time, and a small sketch of the expected Moon shape.

Materials per pair

- 1 printed page of the current month’s *pañcāṅga* (teacher selects ONE source — Drik Panchang, Mypanchang, or any standard *pañcāṅga* — and uses it consistently across the class)
- 1 blank monthly calendar grid (printed handout — see below)
- 2 coloured pencils (yellow for *śukla*, blue/grey for *kṛṣṇa*)
- 1 black pen each

Set-up (teacher, before class)

- Print the *pañcāṅga* page for the current calendar month. One per pair.
- Print the blank calendar grid (template below). One per pair.
- Write the class’s *pañcāṅga* source on the board so every pair uses the same one.

Instructions (student-facing — printable handout)

Build a Tithi Calendar — work in pairs.

1. Look at the *pañcāṅga* page for this month. Find today’s date row.
2. Note the **tithi at sunrise** for today. Find the same tithi on your blank calendar grid for today.

3. Note the **tithi end-time**. After this time, the next tithi begins.
4. Repeat for every date in the month.
5. Colour-code: yellow for *śukla pakṣa* days, blue/grey for *kṛṣṇa pakṣa* days.
6. Circle any *amāvasyā* or *pūrṇimā*.
7. Sketch a small Moon shape next to each date — or every 3rd date if running short.
8. **At the bottom of the calendar, write:** the date of *amāvasyā*, the date of *pūrṇimā*, and any festival you can spot in this month.

Blank calendar grid (one per pair)

Month: _____ Source: _____

Date	Day	Pakṣa	Tithi	Ends at	Moon shape
1					
2					
3					
4					
5					
6					
7					
8					
9					
10					
11					
12					
13					
14					
15					
16					
17					
18					
19					
20					
21					
22					
23					
24					
25					
26					
27					
28					
29					
30					
31					

THIS MONTH:

- Amāvasyā falls on: _____
- Pūrṇimā falls on: _____
- Festival(s) I noticed: _____

Discussion prompts (last 3 min, whole class)

- Which date was the trickiest to figure out? Why?
- Did anyone find a tithi that “skipped a sunrise” (*kṣaya*)? Where?
- How does your calendar compare to the standard Gregorian calendar — what does it tell you that the Gregorian doesn’t?

Looks-like / sounds-like

	Looks like	Sounds like
Going well	Both partners reading the <i>pañcāṅga</i> , one calling out, one writing; colour-coding consistent	“OK, the 14th says caturdaśī, that’s still kṛṣṇa pakṣa, so blue”
Needs intervention	One partner doing all the work; rows skipped; mixed-up colours	“I’ll just write whatever” / “do we have to do every row?”

Safety / sensitivity

- No safety concerns specific to this activity.
- If a student is fasting on a tithi during class (some students observe *ekādaśī* fasts), be aware but don’t single them out.

Differentiation

- **Tier up:** Compute one tithi from scratch using the elongation rule. Use an astronomy app to find the Sun’s and Moon’s ecliptic longitudes for a moment today. Subtract, mod 360, divide by 12, floor, +1 — that’s the tithi number. Compare with what the *pañcāṅga* says.
- **Tier down:** Fill in the date column, day-of-week column, and tithi column only. Skip the end-times and Moon shapes for this lesson.

Clean-up

- Each pair keeps their completed calendar (paste into student workbook on Day 4 page).
- Teacher collects one duplicate copy for the project record.

Activity 1 — 10-Day Moon Observation Diary

Module: Moon Phases and Tithi · **Band:** Middle · **Duration:** 30 min

Purpose

Reinforce the day-by-day concepts of Moon Phases and Tithi through a hands-on activity. Aligned with NEP 2020 §4.23 (experiential learning).

Materials

notebook, pencils, ~15 minutes per evening

What students do

Each evening for 10 days, students sketch the moon shape they see, note the time, the tithi name, and one feeling/observation. On Day 10 they compare sketches with the actual calendar.

Step-by-step

1. **Setup (5 min)** — gather materials; explain the task.
2. **Working phase (20 min)** — students work individually or in pairs.
3. **Share (5 min)** — selected students share their results with the class.

Differentiation

- **Extension:** ask advanced students to do an additional iteration or analysis.
- **Support:** pair students who need scaffolds; offer partial templates.

Assessment

This activity is **formative**. The teacher observes participation, accuracy, and reflection quality. No grade is assigned.

Safety

- General safety briefing before any materials are handled.
- If herbs, plants, or food are involved: kitchen-amount only, no ingestion of unknown preparations.
- Outdoor activities: stay in the designated area; supervised by the teacher.

Activity 02 — Diagram the Eclipse Geometry

Module: Moon Phases & Tithi · **Band:** Middle · **Day:** 7 · **Time:** 10 min (within Day 7's 45 min)

Type

Diagram drawing + short writing, paired

Goal

Each pair draws the Sun–Earth–Moon geometry for both eclipse types, marks the Moon's tilted orbit and its two nodes, and writes a 3-sentence explanation of why eclipses don't happen every month.

Materials per pair

- 1 printed handout (two partial diagrams — see below)
- 2 different colour pencils (one for the ecliptic plane, one for the Moon's tilted orbit)
- 1 black pen each

Set-up (teacher, before class)

- Print the handout. One per pair.
- Have a globe and a small dark ball ready for demo as students work.

Instructions (student-facing — printable handout)

Eclipse Geometry — work in pairs.

Below are two unfinished diagrams. Complete them.

Diagram 1: Solar Eclipse

The Sun is on the left. Earth is on the right. The Moon's orbit around Earth is drawn as a circle around Earth.

1. Place the Moon **between** the Sun and Earth. (Mark it with a small filled circle.)
2. Which tithi must this be? Label it: _____.
3. Draw the Moon's shadow as two lines extending from the Moon to Earth. Shade the region of Earth that falls in the shadow.
4. Below the diagram, write: *"On this tithi, the Moon's shadow covers a small region of Earth. People in this region see a solar eclipse."*

[BLANK SPACE FOR DIAGRAM 1]

Diagram 2: Lunar Eclipse

The Sun is on the left. Earth is in the middle. The Moon's orbit is around Earth.

1. Place the Moon **opposite** the Sun, on the right of Earth. (Mark it with a small open circle.)
2. Which tithi must this be? Label it: _____.
3. Draw Earth's shadow as two lines extending from Earth outward, away from the Sun. The Moon falls inside the shadow.
4. Below the diagram, write: *"On this tithi, the Moon passes through Earth's shadow. Anyone on the night side of Earth can see a lunar eclipse."*

[BLANK SPACE FOR DIAGRAM 2]

Bottom of the page

5. Now draw the Moon's orbit around Earth as a circle **TILTED** by 5° away from the Sun-Earth line. Mark the two **nodes** — the two points where the tilted orbit crosses the Sun-Earth line — with stars.
6. **In 3 sentences, answer:** Why don't eclipses happen every *amāvasyā* and every *pūrṇimā*?

Discussion prompts (whole class, 2 min after activity)

- Show your tilted-orbit diagram. Where on the orbit can eclipses happen? (Only at or near the nodes.)
- About how often does the Moon pass through a node? (Twice each month — but only twice each YEAR does the node alignment land near an *amāvasyā* or *pūrṇimā* — these are the “eclipse seasons.”)

Looks-like / sounds-like

	Looks like	Sounds like
Going well	Both partners drawing carefully; shadow cones drawn at sensible angles; explanation in own words	“OK so if the orbit is tilted, the Moon is usually above or below the Earth’s shadow line”
Needs intervention	Both drawing the same circle on top of each other; explanation copied from textbook	Silence, or “I don’t get it”

Safety / sensitivity

- This is a paper activity. No safety concerns.
- Eclipse-viewing safety is taught in the lesson, not in this activity.

Differentiation

- **Tier up:** Add a third diagram showing the Moon at *amāvasyā* but FAR from a node — show how the Moon’s shadow misses Earth (passes above or below). This is the visual answer to “why not every month.”
- **Tier down:** Just complete Diagram 1 (solar) and Diagram 2 (lunar). Skip the tilted-orbit + nodes detail; the *concept* of “Moon’s orbit is tilted” is enough.

Clean-up

- Handouts paste into the student workbook (Day 7 page).
- Spare paper recycled.

Project Brief

Project Brief — Module 6 Moon Phases & Tithi (Middle)

Work sessions: Day 8 + Day 9 (in class) · **Presentation + summative:** Day 10 · **Total marks:** 40

The project

Each pair (or individual, with teacher approval) chooses **one** lunar phenomenon and produces a small project that shows:

1. **What the *pañcāṅga* / traditional Indian astronomy says** about this phenomenon. Must cite at least one source — the *Sūrya-siddhānta* (paraphrased), the class *pañcāṅga*, NCERT Physics XI, or NASA’s published phase data.
2. **One modern astronomical account** of the same phenomenon — the underlying geometry, mechanism, or measurement.
3. **One personal observation or local example** — a date from your Moon diary, a festival from your family, a local *pañcāṅga* page from your area, etc.

Choose ONE lunar phenomenon

- **Moon phases** — the 8 named phases and the geometry that produces them.
- **Eclipses** — solar, lunar, why they happen at *amāvasyā* / *pūrṇimā*, and the 5° tilt.
- **Tithi calendar** — explain the tithi system, walk through a month’s calendar.
- **Lunar months and festivals** — name the 12 months, link to festivals.
- ***Adhika māsa*** — the intercalary month and why a lunisolar calendar needs it.
- ***Pañcāṅga*** — the five limbs and what each tracks.
- **Cross-cultural lunar calendars** — Indian compared with Islamic / Chinese / Hebrew.

Why this matters

You’re learning to hold two intellectual frameworks side-by-side — the *pañcāṅga* and modern astronomy. They are both real, both useful, and they answer slightly different questions about the same Moon. The discipline of treating both fairly is the deeper learning.

What success looks like

A strong project will: - Demonstrate the module’s central concepts accurately (tithi, pakṣa, *amāvasyā*/*pūrṇimā*, the geometry). - Cite a real source — no inventions. - Connect to a modern astronomical account fairly — without claiming “ancient Indians knew this already.” - Use a real observation from your own diary or calendar. - Be clearly communicated in your chosen medium.

Format options

Format	Best for topics	Notes
Poster (A2 or larger)	Any	Lots of visual space; works for nervous presenters
Live demonstration (2 min hands-on)	Phases (torch and ball), eclipses (shadow geometry)	Needs setup
Short written story (1 page)	Adhika māsa, cross-cultural	Read aloud or hand out
90-second video	Any	Pre-made and played in class
Slide deck (5–8 slides)	Pañcāṅga, calendars	Project from laptop
Infographic	Tithi calendar, month-festival mapping	Combine drawing + facts

Other formats with teacher approval. Whatever the format, the 3-minute time limit on Day 10 applies.

Working in pairs (or individually)

- **Pair work is encouraged**, especially for video/demo formats.
- **Individual is allowed** — fine for a single-topic explainer.
- Groups of 3 are acceptable but each member must have a distinct sub-role.

Schedule

Day	What you do	Submit at end of class
5	Choose topic + format + partner	Sticky note on the board
6	(Homework) Submit project plan paragraph	Project plan paragraph
8	Plan + start first artefact	Draft of first artefact (photo or scan accepted)
9	Refine + rehearse	Draft of final artefact
10	Present (3 min) + Q&A (1 min)	Final artefact + reflection slip

Citation rules

- At minimum **one** citation must appear in your project — visible on the poster, named in the video, or printed in the story.
- The citation must be from a real source: paraphrase of *Sūrya-siddhānta* (with chapter number — chapter 2 for tithi, chapters 4–5 for eclipses, chapter 14 for *adhika māsa*),

NCERT Physics XI, NASA’s official lunar phase publications, the class *pañcāṅga* source.

- **Do not invent specific verse numbers.** If you don’t have a verifiable verse, paraphrase and cite the text by name. Inventing a citation scores 0 on **Use of sources** and on **Originality**.

Rubric

See `rubric.md`. Read it before Day 8.

Self-reflection (submitted on Day 10 with project)

In 4–6 sentences, answer:

1. What did you set out to learn or make, and what did you actually end up with?
2. What is the most surprising thing you discovered while making this?
3. If you had three more days, what would you do differently?

Exemplars

A “Meets” exemplar will be added to `exports/pdf/middle-module-06-moon-phases-tithi/exemplar-strong.pdf` after the first cohort presents.

Rubric

Project Rubric — Module 6 Moon Phases & Tithi (Middle)

Project marks: 30 (out of 40 total when combined with presentation 10).

Criteria

Criterion	4 — Exceeds	3 — Meets	2 — Approaches	1 — Beginning	Weight
Conceptual accuracy	Names the lunar phenomenon AND its astronomical mechanism AND its position in the tithi/pakṣa system correctly. Connects to at least one other	Names the phenomenon and the mechanism, and one accurate fact about its place in the framework. Minor imprecision	Names the phenomenon but mixes up the mechanism OR the position in the framework.	Names the phenomenon only; gets the mechanism or position wrong.	× 2

Criterion	4 — Exceeds	3 — Meets	2 — Approaches	1 — Beginning	Weight
	concept from the module (e.g. ties phases to the elongation rule).	OK.			
Use of sources	Cites at least ONE real source: a paraphrase of the <i>Sūrya-siddhānta</i> (with chapter reference), or a specific reference to NCERT Physics XI, or the class <i>pañcāṅga</i> , or NASA’s published phase data. Bonus: cites two sources.	Cites a real source by name. The source matches a reference in sources .md.	Mentions a source vaguely (“Indian astronomy texts say...” or “scientists have shown...”) without a specific reference.	No source cited, or an invented citation.	× 2
Modern-traditional comparison	Explicitly contrasts the traditional Indian account (tithi, <i>pañcāṅga</i>) with the modern astronomical account, AND names a way each one captures something the other does not.	Names both the traditional and the modern account; explains how they overlap.	Mentions both but doesn’t explain the relationship.	Only one of the two accounts present.	× 1
Use of own observation	Uses the 10-night Moon diary directly	Mentions the diary or a personal	The project is generic; no personal	No observation; clearly from textbook/web	× 1

Criterion	4 — Exceeds	3 — Meets	2 — Approaches	1 — Beginning	Weight
	— for example, shows the diary as data, or refers to specific nights' sketches.	observation .	observation visible.	only.	
Originality of expression	The project shows the pair's own thinking — a turn of phrase, a fresh comparison, a surprising example or angle.	Clearly the students' own work; not lifted from any single source.	Largely paraphrased from a single source; little independent thinking visible.	Clearly copied from a single source.	× 1
Communication / craft	Visuals, layout, or narrative are clean, deliberate, and serve the message. Spelling and IAST are correct.	Communicates clearly. One or two craft issues (a typo, a misaligned heading).	Message comes through despite some craft issues.	Hard to follow; multiple craft issues impede understanding.	× 1

Score calculation

Project score = (sum of criterion scores × weights) × 0.9375

Maximum raw score: 32. Multiply by 30/32 = 0.9375 to get out of 30.

Presentation marks (10, on Day 10)

Criterion	4	3	2	1	Weight
Within time (3 min)	Within 2:45–3:00	Within 2:30–3:15	Within 2:15–3:30	Outside that range	× 1
Both partners contribute (or solo presents whole)	Equal contribution; smooth	Both partners speak	One partner dominates	One partner silent	× 1

Criterion	4	3	2	1	Weight
	h handof fs	but unequ ally			
Clarity and audibility	Confid ent, audibl e to the back of the room, clear pace	Audibl e, mostly clear	Audibl e only in front rows; some mumbl ing	Mostly inaudi ble	× 0.5

Maximum: 10.

Self-reflection mark (folded into project score)

The 3 reflection questions are not separately scored — but if the reflection is missing or perfunctory, the **Originality** mark drops by 1.

Common decision points

- **Group of 3 vs pair:** Acceptable, but the third member must have a clearly distinct sub-role (e.g. visuals + script + presenter). Marks are equal across the group unless one member visibly didn't contribute.
- **Late submission:** -2 marks per school day, capped at -6.
- **Invented citations:** If a project cites a source that doesn't exist (a fake *Sūrya-siddhānta* verse number, a fabricated NASA URL), the project goes to 0 on **Use of sources** AND on **Originality**. A 1-day extension is offered for a redo.
- **Diary incomplete:** If the Moon diary has fewer than 7 nights, the **Use of own observation** mark caps at 2. If under 4 nights, caps at 1.

Marking process

- Mark in real-time during presentation (5 minutes per pair: 3 present + 1 Q&A + 1 marking).
- Write 1-sentence written feedback after each presentation finishes.
- Return marked rubrics within 2 school days.